

第6章 – 船舶与无人水面艇的操纵模型

对稳航形式主义（第5章）的一种替代方法是使用机动理论来描述船舶和无人水面艇的运动。在经典机动理论中，频率依赖的附加质量和辐射阻尼矩阵被常数水动力系数取代，这些系数通常选择为水平面运动（纵摇、横摇和偏航方向）下的零频值。因此，流体记忆效应被忽略，方程可以直接在船体固定参考系中进行形式化。

对于完全耦合的六自由度机动模型，可以通过对频率相关的海况系数进行基于功率的谱平均，来获得恒定的等效附加质量和阻尼矩阵。这样可以得到一个物理一致的一阶原理非线性质量-阻尼-弹簧系统，适用于仿真、制导、导航和控制应用。



章节目标

- 通过将刚体动力学与流体动力附加质量、阻尼、静水恢复力和外力相结合，制定基于第一性原理的六自由度机动模型。
- 解释刚体和附加质量科里奥利-向心矩阵在机体固定坐标系中是如何产生的，并说明它们的斜对称性特性。
- 解释在纵荡、横荡和偏航中何时适合使用零频潜势系数，为什么必须添加粘性阻尼，以及为什么该近似在耦合6自由度运动中失效。
- 应用基于功率的谱平均方法来计算6自由度机动模型的海况相关恒定等效附加质量和阻尼矩阵。
- 使用 ITTC 冲程阻力、横向流拖力以及额外的横摇、纵倾和偏航阻尼来建模线性和非线性流体阻尼。
- 使用相对速度包含恒定无旋转的海洋洋流，并用相对速度或绝对速度作为状态来制定动力学。
- 推导并比较简化和经典的机动模型，包括纵横向、水平面三自由度、动态定位、Abkowitz、二阶模量和MMG模型。

$$\underbrace{M_{RB}\dot{\nu} + C_{RB}(\nu)\nu}_{\text{rigid-body forces}} + \underbrace{M_A\dot{\nu}_r + C_A(\nu_r)\nu_r + D(\nu_r)\nu_r + d(\nu_r)}_{\text{hydrodynamic forces}} + \underbrace{g(\eta)}_{\text{hydrostatic forces}} = \tau + \tau_{\text{wind}} + \tau_{\text{wave}}$$

基于第一性原理的六 自由度机动模型

第6章总结：基于第一性原理的六自由度机动模型

包括洋流的机动模型

$$\underbrace{M_{RB}\dot{\boldsymbol{\nu}} + C_{RB}(\boldsymbol{\nu})\boldsymbol{\nu}}_{\text{rigid-body forces}} + \underbrace{M_A\dot{\boldsymbol{\nu}}_r + C_A(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + D(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{d}(\boldsymbol{\nu}_r)}_{\text{hydrodynamic forces}} + \underbrace{\mathbf{g}(\boldsymbol{\eta})}_{\text{hydrostatic forces}} = \boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}}$$

!c !

$M = M_{RB} + M_A$ – system inertia matrix (including added mass)

$C(\boldsymbol{\nu}_r) = C_{RB}(\boldsymbol{\nu}_r) + C_A(\boldsymbol{\nu}_r)$ – Coriolis-centripetal matrix (including added mass)

$D(\boldsymbol{\nu}_r)$ – hydrodynamic damping matrix

$\mathbf{d}(\boldsymbol{\nu}_r)$ – nonlinear dissipative forces and moments not expressible in the form $D(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r$

$\mathbf{g}(\boldsymbol{\eta})$ – vector of gravitational and buoyancy forces and moments

$\boldsymbol{\tau}$ – vector of control forces and moments

$\boldsymbol{\tau}_{\text{wind}}$ – vector of wind-induced forces and moments

$\boldsymbol{\tau}_{\text{wave}}$ – vector of wave-induced forces and moments

属性 10.1（无旋转的恒定海流）操纵模型可以完全用相对速度向量 $\boldsymbol{\nu}_r = \boldsymbol{\nu} - \boldsymbol{\nu}_c$ 表示。如果刚体科里奥利力和向心力矩阵 $C_{RB}(\boldsymbol{\nu})$ 独立于线速度 $\boldsymbol{\nu}_1 = u, v, w^T$ 参数化（第 3.3.2 节），并且海流

$$[\quad]$$

$$\boldsymbol{\nu}_c = [\underbrace{u_c, v_c, w_c}_{\boldsymbol{\nu}_c^b}, 0, 0, 0]^T$$

是无旋的并且恒定。因此，

$$M_{RB}\dot{\boldsymbol{\nu}} + C_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} = M_{RB}\dot{\boldsymbol{\nu}}_r + C_{RB}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r$$

这意味着

$$M = M_{RB} + M_A$$

$$C(\boldsymbol{\nu}_r) = C_{RB}(\boldsymbol{\nu}_r) + C_A(\boldsymbol{\nu}_r)$$

$$M\dot{\boldsymbol{\nu}}_r + C(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + D(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{d}(\boldsymbol{\nu}_r) + \mathbf{g}(\boldsymbol{\eta}) = \boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}}$$

第6章总结：基于第一性原理的六自由度机动模型

通常使用两种状态空间公式。第一种使用相对速度 $\boldsymbol{\nu}_r$ 作为状态，在明确建模洋流且测量相对速度（例如，通过 DVL）时很方便。第二种使用绝对速度 $\boldsymbol{\nu}$ 作为状态，在表面船舶中通常更受欢迎，因为 INS/GNSS 系统提供绝对速度的测量。

Relative-velocity state-space model:

$$\dot{\boldsymbol{\eta}} = \mathbf{J}_{\boldsymbol{\Theta}}(\boldsymbol{\eta}) (\boldsymbol{\nu}_r + \boldsymbol{\nu}_c)$$

$$\dot{\boldsymbol{\nu}}_r = \mathbf{M}^{-1} (\boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}} - \mathbf{C}(\boldsymbol{\nu}_r) \boldsymbol{\nu}_r - \mathbf{D}(\boldsymbol{\nu}_r) \boldsymbol{\nu}_r - \mathbf{d}(\boldsymbol{\nu}_r) - \mathbf{g}(\boldsymbol{\eta}))$$

$$\boldsymbol{\nu}_c = \underbrace{[u_c, v_c, w_c]^\top}_{\mathbf{v}_c^b}, \quad 0, 0, 0]^\top$$

$$\boldsymbol{\nu}_r = \boldsymbol{\nu} - \boldsymbol{\nu}_c$$

Absolute-velocity state-space model:

$$\dot{\boldsymbol{\eta}} = \mathbf{J}_{\boldsymbol{\Theta}}(\boldsymbol{\eta}) \boldsymbol{\nu}$$

$$\dot{\boldsymbol{\nu}} = \dot{\boldsymbol{\nu}}_c + \mathbf{M}^{-1} (\boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}} - \mathbf{C}(\boldsymbol{\nu}_r) \boldsymbol{\nu}_r - \mathbf{D}(\boldsymbol{\nu}_r) \boldsymbol{\nu}_r - \mathbf{d}(\boldsymbol{\nu}_r) - \mathbf{g}(\boldsymbol{\eta}))$$

$$\mathbf{v}_c^b = [u_c, v_c, w_c]^\top$$

$$\mathbf{v}_c^n = \mathbf{R}(\boldsymbol{\Theta}_{nb}) \mathbf{v}_c^b$$

$$\dot{\boldsymbol{\nu}}_c = \begin{bmatrix} -\mathbf{S}(\boldsymbol{\omega}_{nb}^b) \mathbf{v}_c^b \\ \mathbf{0}_{3 \times 1} \end{bmatrix}$$

Definition 10.1 (Irrotational constant ocean current)

An irrotational ocean current that is constant in the inertial frame $\{n\}$ satisfies

$$\dot{\mathbf{v}}_c^n = \dot{\mathbf{R}}(\boldsymbol{\Theta}_{nb}) \mathbf{v}_c^b + \mathbf{R}(\boldsymbol{\Theta}_{nb}) \dot{\mathbf{v}}_c^b := \mathbf{0}$$

where $\dot{\mathbf{R}}(\boldsymbol{\Theta}_{nb}) = \mathbf{R}(\boldsymbol{\Theta}_{nb}) \mathbf{S}(\boldsymbol{\omega}_{nb}^b)$. Consequently,

$$\dot{\mathbf{v}}_c^b = -\mathbf{S}(\boldsymbol{\omega}_{nb}^b) \mathbf{v}_c^b.$$

刚体动力学

牛顿-欧拉方程 (第3章)

$$\mathbf{M}_{RB}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} = \boldsymbol{\tau} + \boldsymbol{\tau}_{\text{hyd}} + \boldsymbol{\tau}_{\text{hs}} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}}$$

$$\mathbf{M}_{RB} = \begin{bmatrix} m\mathbf{I}_3 & -m\mathbf{S}(\mathbf{r}_{bG}^b) \\ m\mathbf{S}(\mathbf{r}_{bG}^b) & \mathbf{I}_b^b \end{bmatrix}$$

$$= \begin{bmatrix} m & 0 & 0 & 0 & mz_G & -my_G \\ 0 & m & 0 & -mz_G & 0 & mx_G \\ 0 & 0 & m & my_G & -mx_G & 0 \\ 0 & -mz_G & my_G & I_x & -I_{xy} & -I_{xz} \\ mz_G & 0 & -mx_G & -I_{yx} & I_y & -I_{yz} \\ -my_G & mx_G & 0 & -I_{zx} & -I_{zy} & I_z \end{bmatrix}$$

Linear velocity-independent parametrization.

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \mathbf{H}^\top(\mathbf{r}_{bG}^b) \begin{bmatrix} m\mathbf{S}(\boldsymbol{\nu}_2) & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & -\mathbf{S}(\mathbf{I}_b^b \boldsymbol{\nu}_2) \end{bmatrix} \mathbf{H}(\mathbf{r}_{bG}^b)$$

$$= \mathbf{H}^\top(\mathbf{r}_{bG}^b) \begin{bmatrix} 0 & -mr & mq & 0 & 0 & 0 \\ mr & 0 & -mp & 0 & 0 & 0 \\ -mq & mp & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & I_z r & -I_y q \\ 0 & 0 & 0 & -I_z r & 0 & I_x p \\ 0 & 0 & 0 & I_y q & -I_x p & 0 \end{bmatrix} \mathbf{H}(\mathbf{r}_{bG}^b)$$

$$\mathbf{v} = [u, v, w, p, q, r]^\top$$

$$\boldsymbol{\nu}_1 := \mathbf{v}_{nb}^b = [u, v, w]^\top$$

$$\boldsymbol{\nu}_2 := \boldsymbol{\omega}_{nb}^b = [p, q, r]^\top$$

$$\mathbf{M}_{RB} = \mathbf{M}_{RB}^\top > 0$$

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = -\mathbf{C}_{RB}^\top(\boldsymbol{\nu})$$

Screw transformation between two points

$$\boldsymbol{\nu}_{nP} = \mathbf{H}(\mathbf{r}_{bP}^b) \boldsymbol{\nu}$$

$$\mathbf{H}(\mathbf{r}_{bP}^b) := \begin{bmatrix} \mathbf{I}_3 & \mathbf{S}^\top(\mathbf{r}_{bP}^b) \\ \mathbf{0}_{3 \times 3} & \mathbf{I}_3 \end{bmatrix}$$

$$\mathbf{H}^\top(\mathbf{r}_{bP}^b) = \begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_{3 \times 3} \\ \mathbf{S}(\mathbf{r}_{bP}^b) & \mathbf{I}_3 \end{bmatrix}$$

附加质量和线性阻尼的水动力导数

The hydrodynamic forces and moments acting on the body are modeled as

$$\boldsymbol{\tau}_{\text{hyd}} = -\mathbf{M}_A \dot{\boldsymbol{\nu}}_r - \mathbf{C}_A(\boldsymbol{\nu}_r) \boldsymbol{\nu}_r - \mathbf{D}(\boldsymbol{\nu}_r) \boldsymbol{\nu}_r - \mathbf{d}(\boldsymbol{\nu}_r)$$

$$\mathbf{M}_A = \mathbf{M}_A^\top \geq 0,$$

$$\mathbf{D} > 0$$

添加的质量矩阵

$$\mathbf{M}_A = - \begin{bmatrix} X_{\dot{u}} & X_{\dot{v}} & X_{\dot{w}} & X_{\dot{p}} & X_{\dot{q}} & X_{\dot{r}} \\ Y_{\dot{u}} & Y_{\dot{v}} & Y_{\dot{w}} & Y_{\dot{p}} & Y_{\dot{q}} & Y_{\dot{r}} \\ Z_{\dot{u}} & Z_{\dot{v}} & Z_{\dot{w}} & Z_{\dot{p}} & Z_{\dot{q}} & Z_{\dot{r}} \\ K_{\dot{u}} & K_{\dot{v}} & K_{\dot{w}} & K_{\dot{p}} & K_{\dot{q}} & K_{\dot{r}} \\ M_{\dot{u}} & M_{\dot{v}} & M_{\dot{w}} & M_{\dot{p}} & M_{\dot{q}} & M_{\dot{r}} \\ N_{\dot{u}} & N_{\dot{v}} & N_{\dot{w}} & N_{\dot{p}} & N_{\dot{q}} & N_{\dot{r}} \end{bmatrix}$$

SNAME (1950年) 水动力导数的符号

$$M_{A,21} = -Y_{\dot{u}}$$

$$Y = -Y_{\dot{u}} \dot{u}, \quad Y_{\dot{u}} := \frac{\partial Y}{\partial \dot{u}}$$

线性阻尼矩阵

$$\mathbf{D} = - \begin{bmatrix} X_u & X_v & X_w & X_p & X_q & X_r \\ Y_u & Y_v & Y_w & Y_p & Y_q & Y_r \\ Z_u & Z_v & Z_w & Z_p & Z_q & Z_r \\ K_u & K_v & K_w & K_p & K_q & K_r \\ M_u & M_v & M_w & M_p & M_q & M_r \\ N_u & N_v & N_w & N_p & N_q & N_r \end{bmatrix}$$

$$D_{66} = -N_r$$

$$N = -N_r r, \quad N_r := \frac{\partial N}{\partial r}$$

从零频振荡系数的估计

三自由度零频近似（经典机动理论）

零频附加质量和阻尼矩阵可用于纵荡、横荡和偏航运动（自由度 1-2-6），因为在高频下辐射消失

频率。注意，当船舶在波浪中振荡时，它会辐射出外向波。

Added mass matrix: $M_{A,ij} \approx A_{ij}(0)$

Linear damping matrix: $D_{ij} \approx B_{ij}(0) + B_{v,ij}$

不应延伸至升降、横滚和纵摇（自由度 3-4-5）：

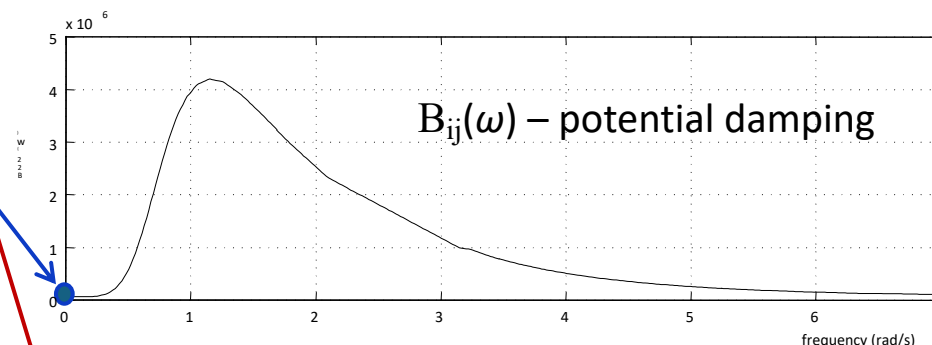
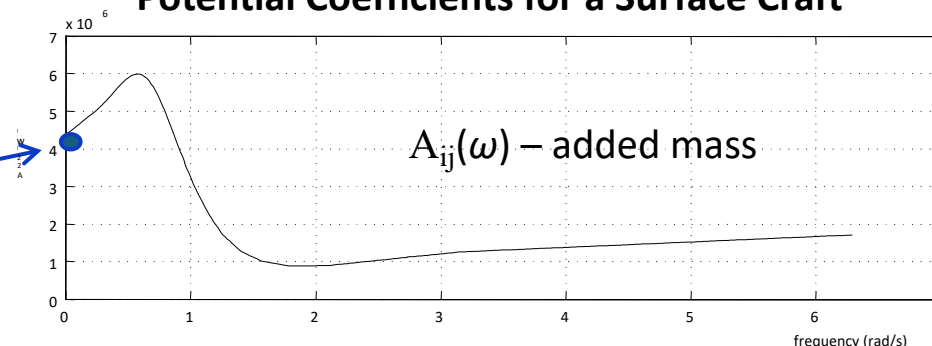
这些运动产生强烈的辐射阻尼，因为垂直方向

振荡会辐射波并耗散能量。它们的固有频率处于波频率范围。因此，系数强烈依赖于频率。仅使用零频率值会导致能量耗散不正确和运动不真实。

耦合项（ A_{13} 、 A_{15} 、 A_{24} 、 A_{26} 、 A_{35} 、 A_{46} ）达到峰值的问题
不同的频率：

o 零频率无法表示这些混合频率效应。 o 解决方法：谱平均，例如基于功率的平均方法（见下一页）。

Potential Coefficients for a Surface Craft

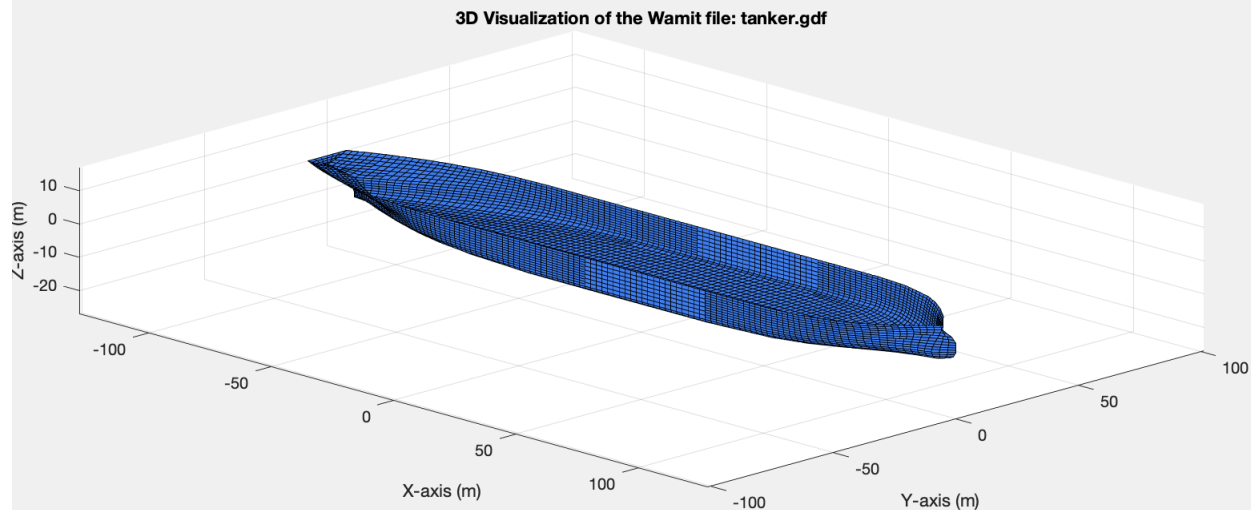


必须添加黏性阻尼 B_v ，因为势能阻尼 $B(0)$ 为零。或者， D 可以通过时间常数/相对阻尼比来估算。

MSS 工具箱

```
>> load tanker
>> vessel
vessel =
  struct with fields:
    main: [1x1 struct]
    velocities: 0
    headings: [1x36 double]
    MRB: [6x6 double]
    C: [6x6x60 double]
    freqs: [1x60 double]
    A: [6x6x60 double]
    B: [6x6x60 double]
    motionRAO: [1x1 struct]
    forceRAO: [1x1 struct]
    driftfr: [1x1 struct]
    Bv: [6x6 double]
>> vessel.freqs
ans =
Columns 1 through 8
    0    0.0383    0.0424    0.0468    0.0517    0.0571    0.0631    0.0697
Columns 9 through 16
    0.0770    0.0850    0.0939    0.1037    0.1145    0.1264    0.1395    0.1540
Columns 17 through 24
    0.1700    0.1876    0.2070    0.2284    0.2519    0.2779    0.3063    0.3378
Columns 25 through 32
    0.3722    0.4101    0.4517    0.4971    0.5473    0.6018    0.6621    0.7281
Columns 33 through 40
    0.7994    0.8775    0.9622    1.0560    1.1571    1.2668    1.3840    1.5104
Columns 41 through 48
    1.6491    1.8003    1.9574    2.1299    2.3100    2.5033    2.7083    2.9361
Columns 49 through 56
    3.1574    3.3963    3.6530    3.9270    4.1888    4.4880    4.7600    5.0671
Columns 57 through 60
    5.3702    5.7120    5.9840    10.0000
```

```
>> plot_wamitgdf('tanker')|
```



```
>> MA = vessel.A(:,:,1)    Zero-frequency added mass matrix
MA =
1.0e+11 *
    0.0000         0    -0.0000         0    -0.0128         0
         0    0.0005         0    0.0030         0    0.0029
   -0.0000         0    0.0039         0    0.0096         0
         0    0.0030         0    0.1103         0   -0.0419
   -0.0128         0    0.0096         0         9.0021         0
         0    0.0029         0   -0.0420         0    1.7383
```

基于功率平均的船舶运动系数的估算

对于一个完全耦合的六自由度机动模型，零频率近似是不合适的。零频率无法表示在不同频率下达到峰值的耦合项 (A_{13} 、 A_{15} 、 A_{24} 、 A_{26} 、 A_{35} 、 A_{46})。然而，频谱平均法可以解决这个问题。

基于功率平均的六自由度等效矩阵 (第5.4节)

频率相关附加质量 $A_U(\omega)$ 和势阻尼 $B_U(\omega)$ 对于恒定速度 U ，在与特定海况相关的整个频率范围内进行积分，并按归一化波谱的谱能量分布加权，以确保只有谱的形状影响等效值 A_U^{eq} 和 B_U^{eq} (Fossen, 2025)：

$$M_A = A_0^{eq}, \quad D = B_0^{eq} + B_v$$

在实际应用中，零阶谱矩 m_0 被评估用于 $U = 0$ ，从而避免了加权谱明确依赖于前进速度 U 和相对波向 β_{rel} ，使得 $\omega_e = \omega$ 。然后，将峰值频率 ω_p 作为描述所考虑操作条件下平均海况的常数参数处理。

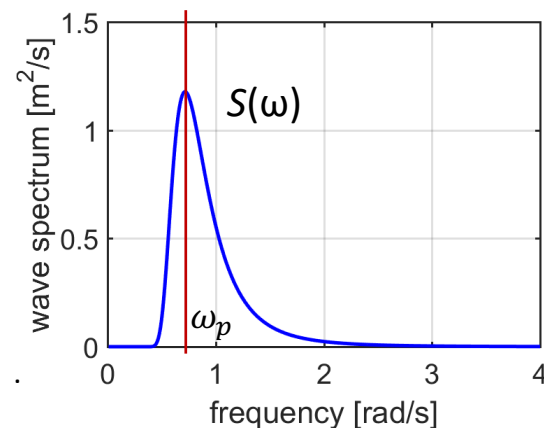
$$A_U^{eq} \approx \frac{1}{m_0} \sum_{k=1}^N A_U(\omega[k]) S(\omega[k]) \Delta\omega$$

$$B_U^{eq} \approx \frac{1}{m_0} \sum_{k=1}^N B_U(\omega[k]) S(\omega[k]) \Delta\omega.$$

用于加权的PM光谱

$$S(\omega) = \frac{\alpha g^2}{\omega^5} \exp\left(-\beta \left(\frac{\omega_p}{\omega}\right)^4\right) \quad S^N(\omega) = \frac{S(\omega)}{m_0}$$

$$m_0 = \int_0^{\omega_{max}} S(\omega) d\omega \approx \sum_{k=1}^N S(\omega[k]) \Delta\omega.$$



Fossen, T. I. (2025). 基于功率方法的频率相关附加质量与阻尼的操纵系数估算
海洋工程 341, 122494. <https://doi.org/10.1016/j.oceaneng.2025.122494>

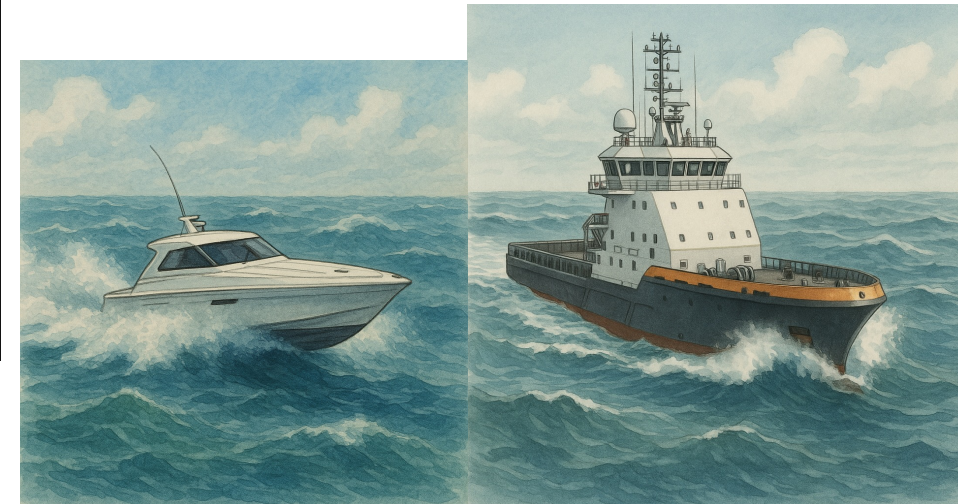
MSS 工具箱

```
load supply;
omega_p = 0.8;
vessel = computeManeuveringModel(vessel, omega_p, 0);
```

```
function vessel = computeManeuveringModel(vessel, omega_p, plotFlag)
% computeManeuveringModel is compatible with MATLAB and GNU Octave (www.octave.org)
% Computes the equivalent added mass vessel.A_eq(:, :, k, n) and damping
% vessel.B_eq(:, :, k, n) matrices for each omega_p(k) and velocity(n) by integrating
% the frequency-dependent hydrodynamic matrices A_U(omega) and B_U(omega) using the wave
% spectrum S(omega) as a weighting function.
%
% The equivalent matrices are computed as:
%
%  $A_{eq} = \int A_U(\omega) S_N(\omega) d\omega$ 
%  $B_{eq} = \int B_U(\omega) S_N(\omega) d\omega$ 
%
% where
%
% S(omega) : Wave energy spectrum
% S_N(omega) = S(omega) / m_0 : Normalized wave spectrum
% m_0 =  $\int S(\omega) d\omega$  : Zero spectral moment
% A_U(omega) and B_U(omega) : Added mass and damping matrices at speed U
%
% The integrals are evaluated numerically using trapezoidal integration.
% This ensures that the kinetic energy and power dissipation properties
% of the frequency-dependent system are preserved in the equivalent
% constant matrices.
```

```
>> vessel.A_eq
ans =
  1.0e+09 *
    0.0014         0         0         0         0         0
         0    0.0047         0    0.0063         0   -0.0084
         0         0    0.0088         0    0.1080         0
         0    0.0063         0    0.0914         0   -0.1839
         0         0    0.1080         0    4.8111         0
         0   -0.0084         0   -0.1839         0    2.0879

>> vessel.B_eq
ans =
  1.0e+09 *
    0.0009         0         0         0         0         0
         0    0.0029         0    0.0038         0   -0.0011
         0         0    0.0061         0    0.0873         0
         0    0.0038         0    0.0197         0   -0.0781
         0         0    0.0873         0    4.1030         0
         0   -0.0011         0   -0.0781         0    0.8871
```



从时间常数和相对阻尼比进行估算

可以用（第6.1.1节）来近似描述海上船舶的响应：

- 时间常数 T_i （一阶系统）。
- 相对阻尼比 ζ_i （二阶系统）。

假设附加质量通过半经验公式，以质心为中心用一个常数对角矩阵来近似：

$$\mathbf{M}_A = \text{diag} \{M_{A,11}, M_{A,22}, M_{A,33}, M_{A,44}, M_{A,55}, M_{A,66}\}$$

$$D_{11} = \frac{m + M_{A,11}}{T_1^*}, \quad D_{22} = \frac{m + M_{A,22}}{T_2^*}, \quad D_{66} = \frac{I_z + M_{A,66}}{T_6^*}.$$

Motion	Time Constant
Surge	$T_1^* = 5\text{--}50 \text{ s}$
Sway	$T_2^* = 1\text{--}20 \text{ s}$
Yaw	$T_6^* = 1\text{--}30 \text{ s}$

$$D_{33} = 2\zeta_3^* \omega_3 (m + M_{A,33})$$

$$D_{44} = 2\zeta_4^* \omega_4 (I_x + M_{A,44})$$

$$D_{55} = 2\zeta_5^* \omega_5 (I_y + M_{A,55})$$

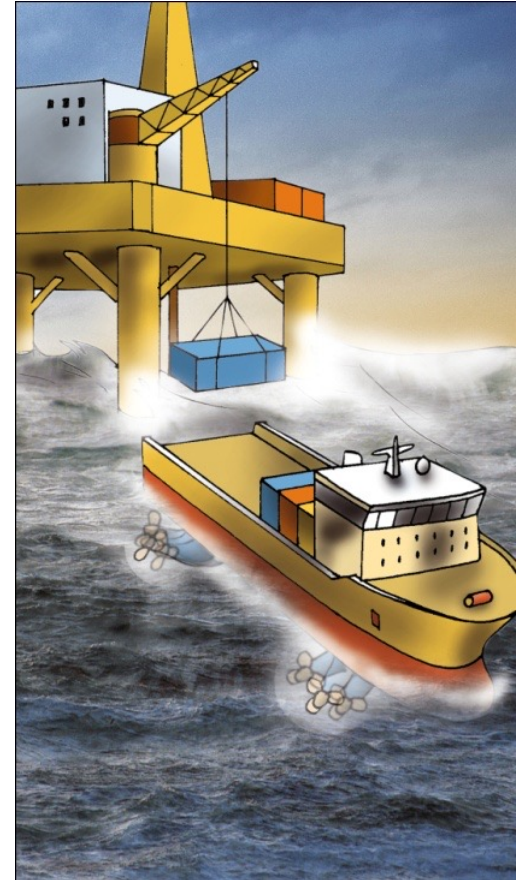
Motion	Relative Damping Ratio
Heave	$\zeta_3^* = 0.1\text{--}0.3$
Roll	$\zeta_4^* = 0.05\text{--}1.0$
Pitch	$\zeta_5^* = 0.1\text{--}0.4$

- Streamlined AUVs and torpedo-shaped vehicles: lower values
- Ships and offshore vessels: higher values

$$\mathbf{D} \approx \begin{bmatrix} D_{11} & 0 & 0 & 0 & 0 & 0 \\ 0 & D_{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & D_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & D_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & D_{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & D_{66} \end{bmatrix}$$

MSS 工具箱

```
function D = Dmtrx(T_126,zeta_45,MRB,MA,hydrostatics)
% D = Dmtrx([T1, T2, T6],[zeta4,zeta5],MRB,MA,hydrostatics)
% computes the 6x6 linear damping matrix for marine craft (submerged and
% floating) by specifying the time constants [T1, T2, T6] in DOFs 1,2 and 6.
% The time constants can be found by open-loop step responses. For roll and
% pitch the relative damping ratios are specified using [zeta4, zeta5].
% For floating vessels it is assumed that zeta3 = 0.2 in heave, while
% submerged vehicles are assumed to be neutrally buoyant,  $W = B$ , with equal
% time constants in heave and sway, that is  $T3 = T2$ .
%
% Inputs: T_126 = [T1, T2, T6]: time constants for DOFs 1, 2 and 6
%         zeta_45 = [zeta4, zeta5]: relative damping ratios in DOFs 4 and 5
%         MRB: 6x6 rigid-body system matrix (see rbody.m)
%         MA: 6x6 hydrodynamic added mass system matrix
%         hydrostatics = G for surface craft (see Gmtrx.m)
%         hydrostatics = [W r_bg' r_bb'] for neutrally buoyant submerged
%         vehicles where  $W = m \cdot g$ ,  $r_{bg} = [xg, yg, zg]'$  and  $r_{bb} = [xb, yb, zb]'$ 
%
% Output: D: 6x6 diagonal linear damping matrix
%
% Examples:
% supply: D = Dmtrx([10 50 20], [0.2, 0.3], MRB, MA, G)
% tanker: D = Dmtrx([20 100 50], [0.2, 0.3], MRB, MA, G)
% AUV:    D = Dmtrx([5 10 10], [0.2, 0.3], MRB, MA, [m*g, [0,0,zg], [0,0,0]])
```



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旋转坐标系中的科里奥利—向心力

Definition 5.1 (Added mass) *Hydrodynamic added mass is the apparent increase in inertia caused by the need to accelerate the surrounding fluid together with the vessel. Hence, an accelerating vessel behaves as if its rigid-body mass and inertia are larger than their physical values.*

In Section 3.1, the rigid-body kinetics of a marine craft were derived using Newton's laws. In an analogous manner, the hydrodynamic added-mass forces and moments can be separated into terms associated with the added-mass system inertia matrix $\mathbf{M}_A$ and terms associated with the added-mass Coriolis–centripetal matrix $\mathbf{C}_A(\boldsymbol{\nu})$. To derive the expression for $\mathbf{C}_A(\boldsymbol{\nu})$, an energy-based approach using Kirchhoff's equations will now be presented.

$\mathbf{C}^A \dot{\mathbf{r}}$

目标是推导出矩阵 $\mathbf{C}_A(\mathbf{v}_r)$ 的公式，以便可以将其作为矩阵 $\mathbf{M}_A = \mathbf{A}_0^{\text{eq}}$ 的函数来计算

$$\underbrace{\mathbf{M}_{RB}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu}}_{\text{rigid-body forces}} + \underbrace{\mathbf{M}_A\dot{\boldsymbol{\nu}}_r + \mathbf{C}_A(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{D}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{d}(\boldsymbol{\nu}_r)}_{\text{hydrodynamic forces}} + \underbrace{\mathbf{g}(\boldsymbol{\eta})}_{\text{hydrostatic forces}} = \boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}}$$

附加质量矩阵 $\mathbf{M}_A$ 不应被解释为随船只携带或拖拽的固定数量的水。这种解释将对应于质量的恒定增加。

相反，附加质量是一种由加速运动产生的压力场引起的动态效应，因此其值取决于物体的几何形状和振动频率。

这个物体的行为就好像它的有效质量是刚体质量和附加质量的总和

$$\mathbf{M} = \mathbf{M}_{RB} + \mathbf{M}_A$$

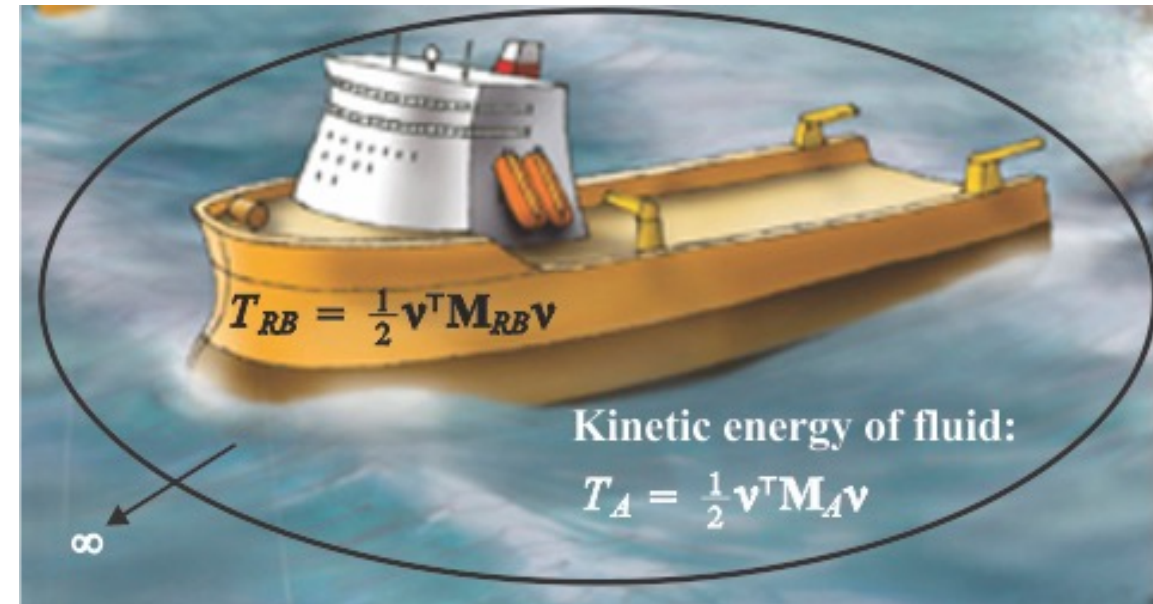
旋转坐标系中的科里奥利-向心力

流体动能

- 流体动能的概念可以用来推导附加质量科氏-向心矩阵
- 容器的任何运动都会引起原本静止的流体发生运动。为了让容器通过流体，它必须向一侧移动，然后在容器之后重新闭合。
- 因此，流体运动具有它本来不会具有的动能（Lamb 1932）。

$$T_A = \frac{1}{2} \boldsymbol{\nu}^T \mathbf{M}_A \boldsymbol{\nu}, \quad \dot{\mathbf{M}}_A = \mathbf{0}$$

$$\mathbf{M}_A = - \begin{bmatrix} X_{\dot{u}} & X_{\dot{v}} & X_{\dot{w}} & X_{\dot{p}} & X_{\dot{q}} & X_{\dot{r}} \\ Y_{\dot{u}} & Y_{\dot{v}} & Y_{\dot{w}} & Y_{\dot{p}} & Y_{\dot{q}} & Y_{\dot{r}} \\ Z_{\dot{u}} & Z_{\dot{v}} & Z_{\dot{w}} & Z_{\dot{p}} & Z_{\dot{q}} & Z_{\dot{r}} \\ K_{\dot{u}} & K_{\dot{v}} & K_{\dot{w}} & K_{\dot{p}} & K_{\dot{q}} & K_{\dot{r}} \\ M_{\dot{u}} & M_{\dot{v}} & M_{\dot{w}} & M_{\dot{p}} & M_{\dot{q}} & M_{\dot{r}} \\ N_{\dot{u}} & N_{\dot{v}} & N_{\dot{w}} & N_{\dot{p}} & N_{\dot{q}} & N_{\dot{r}} \end{bmatrix}$$



周围水体的动能

当船体以速度 $\mathbf{v}$ 移动时，它会在无限域中引发无旋流。考虑一个控制体积，该体积包围船舶并延伸至远场表面 S_∞ （当 $r \rightarrow \infty$ 时水静止）。

该感应流的动能是

$$T_A = \frac{1}{2} \boldsymbol{\nu}^T \mathbf{M}_A \boldsymbol{\nu}$$

所以总动能是（船-水系统）

$$T = \frac{1}{2} \boldsymbol{\nu}^T (\mathbf{M}_{RB} + \mathbf{M}_A) \boldsymbol{\nu}.$$

旋转坐标系中的科里奥利—向心力

可以使用基尔霍夫方程推导矩阵 $\mathbf{C}_A(\mathbf{v}_r)$ 作为矩阵 $\mathbf{M}_A = \mathbf{A}_0^{\text{eq}}$ 的函数的公式。

欧拉-拉格朗日方程（仅适用于广义坐标）

$$L = T - V \quad \text{动能和势能的区别}$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\eta}} \right) - \frac{\partial L}{\partial \eta} = \mathbf{J}_{\Theta}^{-\top}(\eta) \boldsymbol{\tau}$$

基尔霍夫方程

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial T_A}{\partial \dot{\boldsymbol{\nu}}_1} \right) + \mathbf{S}(\boldsymbol{\nu}_2) \frac{\partial T_A}{\partial \boldsymbol{\nu}_1} &= \boldsymbol{\tau}_1 \\ \frac{d}{dt} \left(\frac{\partial T_A}{\partial \dot{\boldsymbol{\nu}}_2} \right) + \mathbf{S}(\boldsymbol{\nu}_2) \frac{\partial T_A}{\partial \boldsymbol{\nu}_2} + \mathbf{S}(\boldsymbol{\nu}_1) \frac{\partial T_A}{\partial \boldsymbol{\nu}_1} &= \boldsymbol{\tau}_2 \end{aligned}$$

6自由度广义坐标：

$$\boldsymbol{\eta} = [x^n, y^n, z^n, \phi, \theta, \psi]^\top$$

四元数不是广义坐标：

$$\boldsymbol{\eta} = [x^n, y^n, z^n, \eta, \varepsilon_1, \varepsilon_2, \varepsilon_3]^\top$$

$$!!! u \, v \, w^\top \quad \quad \quad !!! X \, Y \, Z^\top$$

$$!!! p \, q \, r^\top \quad \quad \quad !!! K \, M \, N^\top$$

$$T_A = \frac{1}{2} \boldsymbol{\nu}^\top \mathbf{M}_A \boldsymbol{\nu}, \quad \dot{\mathbf{M}}_A = \mathbf{0}$$

流体动能

旋转坐标系中的科里奥利一向心力

基尔霍夫方程

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial T_A}{\partial \boldsymbol{\nu}_1} \right) + \mathbf{S}(\boldsymbol{\nu}_2) \frac{\partial T_A}{\partial \boldsymbol{\nu}_1} &= \boldsymbol{\tau}_1 \\ \frac{d}{dt} \left(\frac{\partial T_A}{\partial \boldsymbol{\nu}_2} \right) + \mathbf{S}(\boldsymbol{\nu}_2) \frac{\partial T_A}{\partial \boldsymbol{\nu}_2} + \mathbf{S}(\boldsymbol{\nu}_1) \frac{\partial T_A}{\partial \boldsymbol{\nu}_1} &= \boldsymbol{\tau}_2 \end{aligned}$$

$$T_A = \frac{1}{2} \boldsymbol{\nu}^\top \mathbf{M}_A \boldsymbol{\nu}, \quad \dot{\mathbf{M}}_A = \mathbf{0}$$

Property 6.1 (Added-mass system inertia matrix $\mathbf{M}_A$)

For a rigid body at rest or moving at constant forward speed $U \geq 0$ in an ideal fluid, the added-mass system inertia matrix is symmetric and positive semidefinite:

$$\mathbf{M}_A = \mathbf{M}_A^\top \geq 0.$$

$$\mathbf{M}_A = - \begin{bmatrix} X_{\dot{u}} & X_{\dot{v}} & X_{\dot{w}} & X_{\dot{p}} & X_{\dot{q}} & X_{\dot{r}} \\ Y_{\dot{u}} & Y_{\dot{v}} & Y_{\dot{w}} & Y_{\dot{p}} & Y_{\dot{q}} & Y_{\dot{r}} \\ Z_{\dot{u}} & Z_{\dot{v}} & Z_{\dot{w}} & Z_{\dot{p}} & Z_{\dot{q}} & Z_{\dot{r}} \\ K_{\dot{u}} & K_{\dot{v}} & K_{\dot{w}} & K_{\dot{p}} & K_{\dot{q}} & K_{\dot{r}} \\ M_{\dot{u}} & M_{\dot{v}} & M_{\dot{w}} & M_{\dot{p}} & M_{\dot{q}} & M_{\dot{r}} \\ N_{\dot{u}} & N_{\dot{v}} & N_{\dot{w}} & N_{\dot{p}} & N_{\dot{q}} & N_{\dot{r}} \end{bmatrix}$$

分量式

$$\begin{aligned} X_A &= X_{\dot{u}}\dot{u} + X_{\dot{w}}(\dot{w} + uq) + X_{\dot{q}}\dot{q} + Z_{\dot{w}}wq + Z_{\dot{q}}q^2 \\ &\quad + X_{\dot{v}}\dot{v} + X_{\dot{p}}\dot{p} + X_{\dot{r}}\dot{r} - Y_{\dot{v}}vr - Y_{\dot{p}}rp - Y_{\dot{r}}r^2 \\ &\quad - X_{\dot{v}}ur - Y_{\dot{w}}wr \end{aligned}$$

$$+ Y_{\dot{w}}vq + Z_{\dot{p}}pq - (Y_{\dot{q}} - Z_{\dot{r}})qr$$

$$Y_A = X_{\dot{v}}\dot{u} + Y_{\dot{w}}\dot{w} + Y_{\dot{q}}\dot{q}$$

$$\begin{aligned} &\quad + Y_{\dot{v}}\dot{v} + Y_{\dot{p}}\dot{p} + Y_{\dot{r}}\dot{r} + X_{\dot{v}}vr - Y_{\dot{w}}vp + X_{\dot{r}}r^2 + (X_{\dot{p}} - Z_{\dot{r}})rp - Z_{\dot{p}}p^2 \\ &\quad - X_{\dot{w}}(up - wr) + X_{\dot{u}}ur - Z_{\dot{w}}wp \\ &\quad - Z_{\dot{q}}pq + X_{\dot{q}}qr \end{aligned}$$

$$Z_A = X_{\dot{w}}(\dot{u} - wq) + Z_{\dot{w}}\dot{w} + Z_{\dot{q}}\dot{q} - X_{\dot{u}}uq - X_{\dot{q}}q^2$$

$$\begin{aligned} &\quad + Y_{\dot{w}}\dot{v} + Z_{\dot{p}}\dot{p} + Z_{\dot{r}}\dot{r} + Y_{\dot{v}}vp + Y_{\dot{r}}rp + Y_{\dot{p}}p^2 \\ &\quad + X_{\dot{v}}up + Y_{\dot{w}}wp \end{aligned}$$

$$- X_{\dot{v}}vq - (X_{\dot{p}} - Y_{\dot{q}})pq - X_{\dot{r}}qr$$

$$K_A = X_{\dot{p}}\dot{u} + Z_{\dot{p}}\dot{w} + K_{\dot{q}}\dot{q} - X_{\dot{v}}wu + X_{\dot{r}}uq - Y_{\dot{w}}w^2 - (Y_{\dot{q}} - Z_{\dot{r}})wq + M_{\dot{r}}r^2$$

$$\begin{aligned} &\quad + Y_{\dot{p}}\dot{v} + K_{\dot{p}}\dot{p} + K_{\dot{r}}\dot{r} + Y_{\dot{w}}v^2 - (Y_{\dot{q}} - Z_{\dot{r}})vr + Z_{\dot{p}}vp - M_{\dot{r}}r^2 - K_{\dot{q}}rp \\ &\quad + X_{\dot{w}}uv - (Y_{\dot{v}} - Z_{\dot{w}})vw - (Y_{\dot{r}} + Z_{\dot{q}})wr - Y_{\dot{p}}wp - X_{\dot{q}}ur \\ &\quad + (Y_{\dot{r}} + Z_{\dot{q}})vq + K_{\dot{r}}pq - (M_{\dot{q}} - N_{\dot{r}})qr \end{aligned}$$

$$M_A = X_{\dot{q}}(\dot{u} + wq) + Z_{\dot{q}}(\dot{w} - uq) + M_{\dot{q}}\dot{q} - X_{\dot{w}}(u^2 - w^2) - (Z_{\dot{w}} - X_{\dot{u}})wu$$

$$\begin{aligned} &\quad + Y_{\dot{q}}\dot{v} + K_{\dot{q}}\dot{p} + M_{\dot{r}}\dot{r} + Y_{\dot{p}}vr - Y_{\dot{r}}vp - K_{\dot{r}}(p^2 - r^2) + (K_{\dot{p}} - N_{\dot{r}})rp \\ &\quad - Y_{\dot{w}}uv + X_{\dot{v}}vw - (X_{\dot{r}} + Z_{\dot{p}})(up - wr) + (X_{\dot{p}} - Z_{\dot{r}})(wp + ur) \\ &\quad - M_{\dot{r}}pq + K_{\dot{q}}qr \end{aligned}$$

$$N_A = X_{\dot{r}}\dot{u} + Z_{\dot{r}}\dot{w} + M_{\dot{r}}\dot{q} + X_{\dot{v}}u^2 + Y_{\dot{w}}wu - (X_{\dot{p}} - Y_{\dot{q}})uq - Z_{\dot{p}}wq - K_{\dot{q}}q^2$$

$$\begin{aligned} &\quad + Y_{\dot{r}}\dot{v} + K_{\dot{r}}\dot{p} + N_{\dot{r}}\dot{r} - X_{\dot{v}}v^2 - X_{\dot{r}}vr - (X_{\dot{p}} - Y_{\dot{q}})vp + M_{\dot{r}}rp + K_{\dot{q}}p^2 \\ &\quad - (X_{\dot{u}} - Y_{\dot{v}})uv - X_{\dot{w}}vw + (X_{\dot{q}} + Y_{\dot{p}})up + Y_{\dot{r}}ur + Z_{\dot{q}}wp \\ &\quad - (X_{\dot{q}} + Y_{\dot{p}})vq - (K_{\dot{p}} - M_{\dot{q}})pq - K_{\dot{r}}qr \end{aligned}$$

旋转坐标系中的科里奥利—向心力

附加质量科里奥利和向心矩阵是通过收集基尔霍夫方程中所有不依赖于刚体加速度的项来得到的。

性质 6.2 (附加质量科里奥利-向心矩阵) 对于在理想流体中运动的刚体, 流体动力学的科里奥利和向心矩阵总可以参数化, 使其为斜对称矩阵

$$\mathbf{C}_A(\boldsymbol{\nu}) = -\mathbf{C}_A^\top(\boldsymbol{\nu}), \quad \forall \boldsymbol{\nu} \in \mathbb{R}^6$$

$$\mathbf{C}_A(\boldsymbol{\nu}) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & -\mathbf{S}(\mathbf{A}_{11}\boldsymbol{\nu}_1 + \mathbf{A}_{12}\boldsymbol{\nu}_2) \\ -\mathbf{S}(\mathbf{A}_{11}\boldsymbol{\nu}_1 + \mathbf{A}_{12}\boldsymbol{\nu}_2) & -\mathbf{S}(\mathbf{A}_{21}\boldsymbol{\nu}_1 + \mathbf{A}_{22}\boldsymbol{\nu}_2) \end{bmatrix}$$

$$\mathbf{C}_A(\boldsymbol{\nu}) = \begin{bmatrix} 0 & 0 & 0 & 0 & -a_3 & a_2 \\ 0 & 0 & 0 & a_3 & 0 & -a_1 \\ 0 & 0 & 0 & -a_2 & a_1 & 0 \\ 0 & -a_3 & a_2 & 0 & -b_3 & b_2 \\ a_3 & 0 & -a_1 & b_3 & 0 & -b_1 \\ -a_2 & a_1 & 0 & -b_2 & b_1 & 0 \end{bmatrix}$$

$$a_1 = X_{\dot{u}}u + X_{\dot{v}}v + X_{\dot{w}}w + X_{\dot{p}}p + X_{\dot{q}}q + X_{\dot{r}}r$$

$$a_2 = Y_{\dot{u}}u + Y_{\dot{v}}v + Y_{\dot{w}}w + Y_{\dot{p}}p + Y_{\dot{q}}q + Y_{\dot{r}}r$$

$$a_3 = Z_{\dot{u}}u + Z_{\dot{v}}v + Z_{\dot{w}}w + Z_{\dot{p}}p + Z_{\dot{q}}q + Z_{\dot{r}}r$$

$$b_1 = K_{\dot{u}}u + K_{\dot{v}}v + K_{\dot{w}}w + K_{\dot{p}}p + K_{\dot{q}}q + K_{\dot{r}}r$$

$$b_2 = M_{\dot{u}}u + M_{\dot{v}}v + M_{\dot{w}}w + M_{\dot{p}}p + M_{\dot{q}}q + M_{\dot{r}}r$$

$$b_3 = N_{\dot{u}}u + N_{\dot{v}}v + N_{\dot{w}}w + N_{\dot{p}}p + N_{\dot{q}}q + N_{\dot{r}}r$$

Sagatun, S. I. 和 T. I. Fossen (1991)。水下车辆动力学的拉格朗日公式。 *IEEE International Conference on Systems, Man, and Cybernetics*, 弗吉尼亚州夏洛茨维尔, 美国, IEEE Xplore, 页 1029-1034。

MSS 工具箱

```
function C = m2c(M,nu)
% C = m2c(M,nu) computes the Coriolis-centripetal matrix C(nu) from the
% the system inertia matrix M > 0 for varying velocity nu.
% If M is a 6x6 matrix and nu = [u, v, w, p, q, r]', the output is a 6x6 C matrix
% If M is a 3x3 matrix and nu = [u, v, r]', the output is a 3x3 C matrix.
%
% Examples: CRB = m2c(MRB,nu)
%           CA = m2c(MA, nu)
%
% Output:
%   C: Coriolis-centripetal matrix C = C(nu)
%
% Inputs:
%   M: 6x6 or 3x3 rigid-body MRB or added mass MA system matrix
%   nu: nu = [u, v, w, p, q, r]' or nu = [u, v, r]'
%
% The Coriolis and centripetal matrix depends on nu1 = [u,v,w]' and nu2 =
% [p,q,r]' as shown in Fossen (2021, Theorem 3.2). Alternatively, the matrix
% CRB = CRB(nu2) can be computed using
%
% [MRB,CRB] = rbody(m,R44,R55,R66,nu2,r_bp)

M = 0.5 * (M + M'); % Symmetrization of the inertia matrix

if (length(nu) == 6) % 6-DOF model
    M11 = M(1:3,1:3); M12 = M(1:3,4:6); M21 = M12'; M22 = M(4:6,4:6);

    nu1 = nu(1:3);
    nu2 = nu(4:6);
    nu1_dot = M11 * nu1 + M12 * nu2;
    nu2_dot = M21 * nu1 + M22 * nu2;

    C = [ zeros(3,3)  -Smtrx(nu1_dot)
          -Smtrx(nu1_dot) -Smtrx(nu2_dot) ];

else % 3-DOF model (surge, sway and yaw)
    C = [ 0 0 -M(2,2)*nu(2)-M(2,3)*nu(3)
          0 0 M(1,1)*nu(1)
          M(2,2)*nu(2)+M(2,3)*nu(3) -M(1,1)*nu(1) 0 ];

end
```

Note that C_{RB} and C_A can be computed automatically from M_{RB} and M_A both for 3-DOF and 6-DOF models.

$$M_A = - \begin{bmatrix} X_{\dot{u}} & X_{\dot{v}} & X_{\dot{w}} & X_{\dot{p}} & X_{\dot{q}} & X_{\dot{r}} \\ Y_{\dot{u}} & Y_{\dot{v}} & Y_{\dot{w}} & Y_{\dot{p}} & Y_{\dot{q}} & Y_{\dot{r}} \\ Z_{\dot{u}} & Z_{\dot{v}} & Z_{\dot{w}} & Z_{\dot{p}} & Z_{\dot{q}} & Z_{\dot{r}} \\ K_{\dot{u}} & K_{\dot{v}} & K_{\dot{w}} & K_{\dot{p}} & K_{\dot{q}} & K_{\dot{r}} \\ M_{\dot{u}} & M_{\dot{v}} & M_{\dot{w}} & M_{\dot{p}} & M_{\dot{q}} & M_{\dot{r}} \\ N_{\dot{u}} & N_{\dot{v}} & N_{\dot{w}} & N_{\dot{p}} & N_{\dot{q}} & N_{\dot{r}} \end{bmatrix}$$

$$C_A(\nu) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & -S(A_{11}\nu_1 + A_{12}\nu_2) \\ -S(A_{11}\nu_1 + A_{12}\nu_2) & -S(A_{21}\nu_1 + A_{22}\nu_2) \end{bmatrix}$$

$$M_A = M_A^T = - \begin{bmatrix} X_{\dot{u}} & 0 & 0 \\ 0 & Y_{\dot{v}} & Y_{\dot{r}} \\ 0 & Y_{\dot{r}} & N_{\dot{r}} \end{bmatrix} \quad (N_{\dot{v}} = Y_{\dot{r}})$$

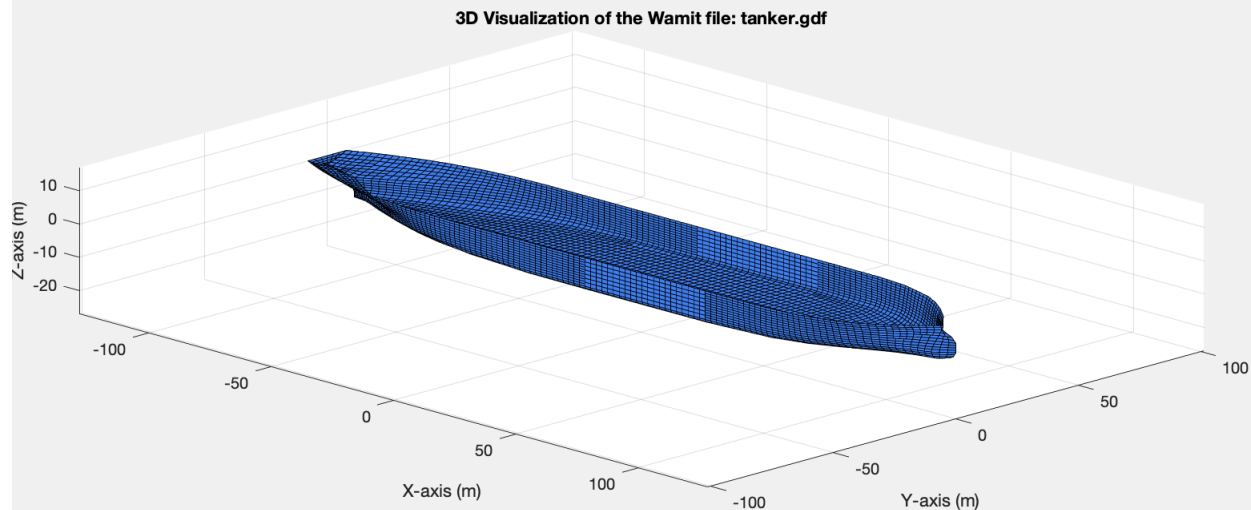
$$C_A(\nu_r) = -C_A^T(\nu_r) = \begin{bmatrix} 0 & 0 & Y_{\dot{v}}v_r + Y_{\dot{r}}r \\ 0 & 0 & -X_{\dot{u}}u_r \\ -Y_{\dot{v}}v_r - Y_{\dot{r}}r & X_{\dot{u}}u_r & 0 \end{bmatrix}$$

MSS 工具箱

```
>> load tanker
>> vessel
vessel =
  struct with fields:

    main: [1x1 struct]
 velocities: 0
  headings: [1x36 double]
    MRB: [6x6 double]
    C: [6x6x60 double]
    freqs: [1x60 double]
    A: [6x6x60 double]
    B: [6x6x60 double]
 motionRAO: [1x1 struct]
 forceRAO: [1x1 struct]
 driftfr: [1x1 struct]
    Bv: [6x6 double]
>> vessel.freqs
ans =
Columns 1 through 8
  0 0.0383 0.0424 0.0468 0.0517 0.0571 0.0631 0.0697
Columns 9 through 16
  0.0770 0.0850 0.0939 0.1037 0.1145 0.1264 0.1395 0.1540
Columns 17 through 24
  0.1700 0.1876 0.2070 0.2284 0.2519 0.2779 0.3063 0.3378
Columns 25 through 32
  0.3722 0.4101 0.4517 0.4971 0.5473 0.6018 0.6621 0.7281
Columns 33 through 40
  0.7994 0.8775 0.9622 1.0560 1.1571 1.2668 1.3840 1.5104
Columns 41 through 48
  1.6491 1.8003 1.9574 2.1299 2.3100 2.5033 2.7083 2.9361
Columns 49 through 56
  3.1574 3.3963 3.6530 3.9270 4.1888 4.4880 4.7600 5.0671
Columns 57 through 60
  5.3702 5.7120 5.9840 10.0000
```

```
>> plot_wamitgdf('tanker')
```



```
>> MA = vessel.A(:,:,1) 零频附加质量矩阵
```

```
MA =
  1.0e+11 *
    0.0000         0 -0.0000         0 -0.0128         0
         0    0.0005         0    0.0030         0    0.0029
   -0.0000         0    0.0039         0    0.0096         0
         0    0.0030         0    0.1103         0   -0.0419
   -0.0128         0    0.0096         0    9.0021         0
         0    0.0029         0   -0.0420         0    1.7383
```

```
>> nu = [10 0 0 0 0 0]';
```

```
>> CA = m2c(MA,nu)
```

```
CA =
  1.0e+10 *
         0         0         0         0 -0.0032         0
         0         0         0    0.0032         0    0.0041
         0         0         0         0 -0.0041         0
         0   -0.0032         0         0         0    1.2758
    0.0032         0    0.0041         0         0         0
         0   -0.0041         0   -1.2758         0         0
```

旋转坐标系中的科里奥利—向心力

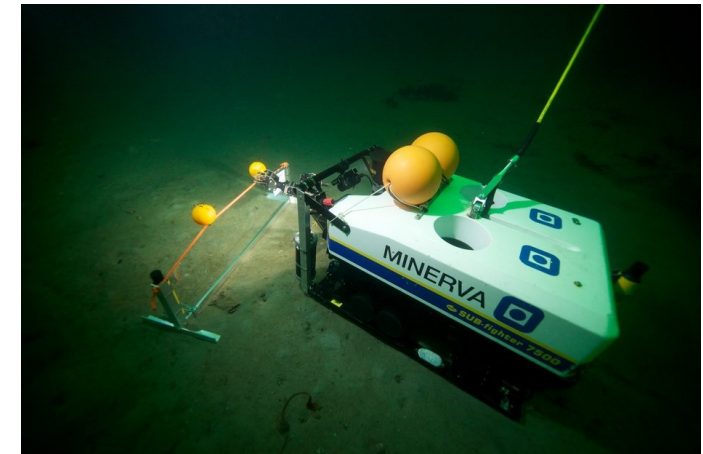
Example 6.2 (Added mass for low-speed AUVs and ROVs)

In general, the motion of an underwater vehicle moving in 6 DOFs at high speed is highly nonlinear and coupled. However, in many AUV and ROV applications, the vehicle operates at low speed. If the vehicle also has three planes of symmetry and the body-fixed frame is located at the CG, the off-diagonal elements of the added-mass matrix M_A are usually small and may therefore be neglected. Hence, the following simplified expressions for M_A and $C_A(\boldsymbol{\nu}_r)$ are obtained:

$$M_A = M_A^\top = -\text{diag}\{X_{\dot{u}}, Y_{\dot{v}}, Z_{\dot{w}}, K_{\dot{p}}, M_{\dot{q}}, N_{\dot{r}}\}$$

$$C_A(\boldsymbol{\nu}_r) = -C_A^\top(\boldsymbol{\nu}_r) = \begin{bmatrix} 0 & 0 & 0 & 0 & -Z_{\dot{w}}w_r & Y_{\dot{v}}v_r \\ 0 & 0 & 0 & Z_{\dot{w}}w_r & 0 & -X_{\dot{u}}u_r \\ 0 & 0 & 0 & -Y_{\dot{v}}v_r & X_{\dot{u}}u_r & 0 \\ 0 & -Z_{\dot{w}}w_r & Y_{\dot{v}}v_r & 0 & -N_{\dot{r}}r & M_{\dot{q}}q \\ Z_{\dot{w}}w_r & 0 & -X_{\dot{u}}u_r & N_{\dot{r}}r & 0 & -K_{\dot{p}}p \\ -Y_{\dot{v}}v_r & X_{\dot{u}}u_r & 0 & -M_{\dot{q}}q & K_{\dot{p}}p & 0 \end{bmatrix}.$$

The diagonal approximation is widely used because it is time consuming to determine the off-diagonal elements experimentally or theoretically. In practice, the approximation is often quite accurate since the off-diagonal elements of a positive inertia matrix are typically much smaller than the corresponding diagonal elements.



旋转坐标系中的科里奥利—向心力

Example 6.1 (Added mass for ships and USVs)

For surface ships and USVs, it is common to decouple the surge mode from the steering dynamics due to symmetry about the xz plane. Similarly, the heave, pitch, and roll modes are neglected under the assumption that these motions are small. Hence, for $\boldsymbol{\nu}_r = [u_r, v_r, r]^\top$ the added-mass matrix and added-mass Coriolis–centripetal matrix become

$$\mathbf{M}_A = \mathbf{M}_A^\top = - \begin{bmatrix} X_{\dot{u}} & 0 & 0 \\ 0 & Y_{\dot{v}} & Y_{\dot{r}} \\ 0 & Y_{\dot{r}} & N_{\dot{r}} \end{bmatrix} \quad (N_{\dot{v}} = Y_{\dot{r}})$$

$$\mathbf{C}_A(\boldsymbol{\nu}_r) = -\mathbf{C}_A^\top(\boldsymbol{\nu}_r) = \begin{bmatrix} 0 & 0 & Y_{\dot{v}}v_r + Y_{\dot{r}}r \\ 0 & 0 & -X_{\dot{u}}u_r \\ -Y_{\dot{v}}v_r - Y_{\dot{r}}r & X_{\dot{u}}u_r & 0 \end{bmatrix}.$$

The Coriolis and centripetal forces corresponding to (6.42) are

$$\mathbf{C}_A(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r = \begin{bmatrix} Y_{\dot{v}}v_r r + Y_{\dot{r}}r^2 \\ -X_{\dot{u}}u_r r \\ \underbrace{(X_{\dot{u}} - Y_{\dot{v}})u_r v_r - Y_{\dot{r}}u_r r}_{\text{Munk moment}} \end{bmatrix}$$

where the first term in the yaw moment is the nonlinear Munk moment, which is known to have destabilizing effects.



非线性阻尼力

势理论代码假设流体是无黏性的。水被假定为无黏性流体，其中流体的黏度等于零。当忽略黏性力时，例如在无黏性流动的情况下，Navier-Stokes 方程可以简化为一种称为欧拉方程的形式。

因此，潜在代码只计算：

ü 势阻尼：无粘流体中的耗散力。与其他耗散项如粘性阻尼相比，势阻尼项的贡献通常可以忽略不计。

流体动力学悖论

达朗贝尔悖论（或称流体力学悖论）由让·勒·朗·达朗贝尔于1752年首次提出，指出在不可压缩、无粘性且无旋的势流中，物体相对于流体以恒定速度运动时所受的阻力为零。这个理论结果与实验观察相矛盾，实验中移动于空气或水等流体中的物体会经历显著的阻力，尤其是在高雷诺数时。

因此，除了势阻尼外，还必须在船舶耐波性和操纵性模型中考虑黏性阻尼力，例如二次阻力和升力。



让·勒·朗·达朗贝尔 (1717-1783)

非线性阻尼力

粘性阻尼的来源

皮肤摩擦

- 船体上的边界层剪应力。
- 对振荡运动影响很小，但总是存在。

涡旋和涡脱落

- 船舷龙骨、船底翼缘、附加装置处的流动分离。
- 对横摇阻尼的主要贡献。

升力引起的阻尼

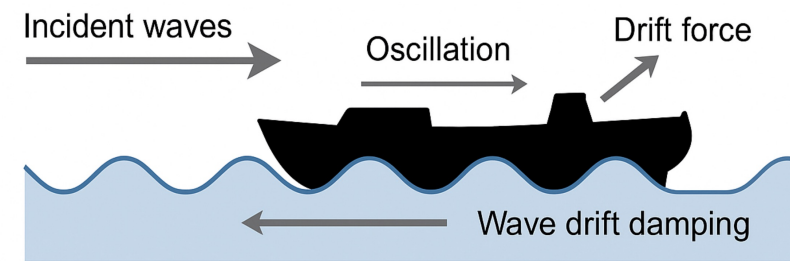
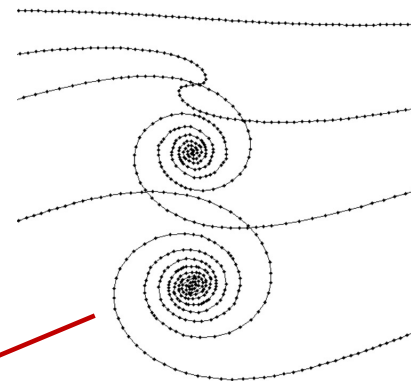
- 产生侧向力的鳍、舵和附属物。
- 对于控制和稳定很重要。

波漂移阻尼

- 平均漂移力与振动的相互作用。
- 通常在二阶波浪载荷分析中处理。

其他非线性粘性效应

- 流动分离、湍流、附属物与船体的相互作用。
- 通常归入经验系数。



- When a vessel oscillates in waves, it experiences order steady drift force (mean wave drift).
- If the vessel is moving relative to this drift, part of the drift force acts as a resisting force : wave drift damping.
- This effect grows with wave height H_s & depends on heading, frequency, and hull form.

Unfortunately, it is very hard to model all these effects and collect the terms into a common nonlinear damping matrix $\mathbf{D}(\mathbf{v})$. Linear damping must be blended with the nonlinear damping terms to avoid “double counting) at higher speed , e.g., using $\mathbf{D} \exp(-\alpha U_g) \mathbf{v} + \mathbf{D}(\mathbf{v}) \mathbf{v}$.

非线性阻尼力

Property 6.3 (Hydrodynamic damping)

The damping matrix $D(\nu_r)$ and the nonlinear damping vector $d(\nu_r)$ satisfy

$$x^T D(\nu_r) x > 0, \quad \forall x \neq 0$$

and

$$\nu_r^T d(\nu_r) \geq 0, \quad \forall \nu_r \neq 0.$$

Hence, the total damping force satisfies

$$\nu_r^T (D(\nu_r) \nu_r + d(\nu_r)) > 0, \quad \forall \nu_r \neq 0$$

so that mechanical energy is always dissipated.

$$\underbrace{M_{RB} \dot{\nu} + C_{RB}(\nu) \nu}_{\text{rigid-body forces}} + \underbrace{M_A \dot{\nu}_r + C_A(\nu_r) \nu_r + D(\nu_r) \nu_r + d(\nu_r)}_{\text{hydrodynamic forces}} + \underbrace{g(\eta)}_{\text{hydrostatic forces}} = \tau + \tau_{\text{wind}} + \tau_{\text{wave}}$$

向量 $d \nu_r$ 是非线性耗散力和力矩，不能以 $D \nu_r \nu_r$ 的形式表示

性质 6.3 适用于线性和非线性阻尼。

线性阻尼

对于像船舶和无人水面艇这样细长的船体，通常会将纵向速度与其他自由度解耦。因此，我们可以将阻尼矩阵 D 表示为水动力导数的函数：

latex sha1_base64"MkEw301C3YdnrS7FYeIQatgYaT0"AAADNnicfZLNbtQwEMedli8SvrZw5GKxAnFhlVSoVKqQqsIBaemqSGy7sFlFjjO7teo4qe20XVl5Ki48B7deOIAQVx4BZ7tFW7cwUjT/mfH87IydlpwpHYannr907fqNm8u3gtt37t6731p5sKuKSILo04IXcpASBZwJ6GumOQxKCSRPOeylB6bt4RSMUK8UFPSxjlZCLYmFGibSpZ8d7FacEzNc2tM29qHG8ijfwexzEKUyYMGIOtGQndTBITFXipzjEjf3P43FwHxMZNhik4船体运动系数或4/pi f179K5D7w07/p2w59yp6z6H3NHcyoQcxiOzykJIWQyEM8OXRTQXbTS3naT1Nc4Kw6阻尼矩阵 D = B^f + B_sMcrrD0SkFJ6AGZwNBKQXJQIzO79ho/sZkMjwtpP6HxLLvYYUiumou1K359pVba5JX1YaVHqPDBNlpUHQs43GEce6wM0bwhmTODWfWkGoZPa

非线性浪涌阻尼

ITTC 阻力

$$X = -\frac{1}{2}\rho S(1+k)C_f(u_r)|u_r|u_r$$

$$C_f(u_r) = \underbrace{\frac{0.075}{(\log_{10}(R_n) - 2)^2}}_{C_F} + C_R$$

哪里

ρ 水的密度 S 船体的湿表面积 k 给出粘性修正的形状因子（对于航行中的船舶通常为 0.1） C_f 来自 ITTC (1957) 曲线的平板摩擦 C_R 由船体粗糙度、压力阻力、波浪阻力引起的剩余摩擦

雷诺数:

$$R_n = \frac{L_{pp}}{\nu}|u_r| \geq 0 \quad \nu = 1 \times 10^{-6} \text{ m}^2/\text{s is the kinematic viscosity at } 20^\circ \text{ C.}$$

SNAME (1950) 的符号

$$X = X_{|u|u}|u_r|u_r$$

$$X_{|u|u} = -\frac{1}{2}\rho S(1+k)C_f(u_r) < 0$$

$$u_r = u - u_c$$

$$= u - V_c \cos(\beta_{V_c} - \psi)$$



对于DP应用，ITTC公式通常给出的阻尼比模型试验中观察到的要少。因此，通常在低速时会加入一个额外的线性阻尼项。在高速时，二次阻力项变得占主导地位，而线性阻尼的贡献应逐渐消失。因此，

$$X = e^{-k_u|u_r|}X_u u_r + X_{|u|u}|u_r|u_r \quad k_u = 3-5,$$

横流阻力原理

对于相对流速角 $|\beta_{vc} - \psi| \gg 0$, 其中 β_{vc} 是洋流方向, 可应用横流阻力原理来计算横摇的非线性阻尼力和偏航力矩 (Faltinsen, 1990)

$$Y = -\frac{\rho C_d^{2D}}{2} \int_{-\frac{L_{pp}}{2}}^{\frac{L_{pp}}{2}} T(x) |v_r + xr| (v_r + xr) dx,$$

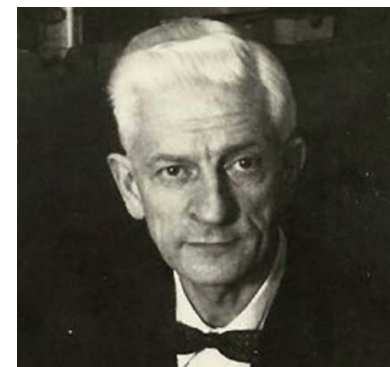
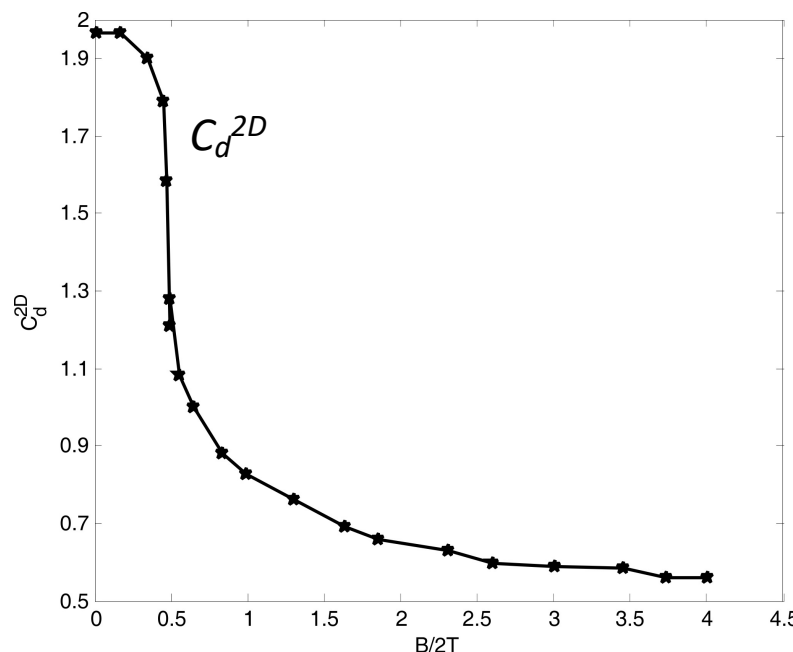
$$N = -\frac{\rho C_d^{2D}}{2} \int_{-\frac{L_{pp}}{2}}^{\frac{L_{pp}}{2}} T(x) x |v_r + xr| (v_r + xr) dx$$

其中 C_d^{2D} 是二维阻力系数, $T(x)$ 是吃水。

条带理论意味着船沿 x 轴被切成若干截面。积分通过将每个截面的贡献相加来计算。

扩展到六自由度横向阻力

$$d(\nu_r) = -\frac{\rho C_d^{2D}}{2} \int_{-\frac{L}{2}}^{\frac{L}{2}} \begin{bmatrix} 0 \\ T(x) |v_r + xr| (v_r + xr) \\ T(x) |w_r + xq| (w_r + xq) \\ 0 \\ x T(x) |w_r + xq| (w_r + xq) \\ x T(x) |v_r + xr| (v_r + xr) \end{bmatrix} dx$$



西加德·F·霍尔纳博士 (1906-1971) 是两本著名空气动力学汇编的作者, 分别名为《流体动力学阻力》和《流体动力学升力》。

Matlab/GNU Octave:

The 2-D drag coefficient C_d^{2D} can be estimated as a function of beam B and draft T using Hoerner's curve. This is implemented in the MSS toolbox by `Cd = Hoerner(B,T)`. The function

```
tau_crossflow = crossFlowDrag(L,B,T,nu_r,drag_model)
```

computes the 6-DOF cross-flow drag integral using 20 strip sections. The default setting is `drag_model = 'Hoerner'`. Alternatively, `'cylinder'` uses the DNV recommended practice for environmental conditions and environmental loads to compute C_d^{2D} for a cylinder-shaped vehicle.

MSS 工具箱

```
>> L = 20; B = 8; T = 5;
>> nu_r = [1 0.1 0.2 0.1 0.1 -0.1]';
>> tau_crossflow = crossFlowDrag(L,B,T,nu_r,'Hoerner')
tau_crossflow =
    1.0e+05 *
         0
    -0.0510
    -0.1030
         0
    -1.4979
     1.4222
>> tau_crossflow = crossFlowDrag(L,B,T,nu_r,'cylinder')
tau_crossflow =
    1.0e+04 *
         0
    -0.3022
    -0.6099
         0
    -8.8728
     8.4242
```

“圆柱体”选项适用于圆柱形AUV，使用 DNV-RP-C 205 的圆柱体数据。而常规船舶船体则更准确地用 Hoerner 曲线表示。

```
function tau_crossflow = crossFlowDrag(L,B,T,nu_r,drag_model)
% tau_crossflow = crossFlowDrag(L,B,T,nu_r,drag_model) computes the cross-flow
% drag integrals for a marine craft using strip theory. Application:
%
%      M * d/dt nu_r + C(nu_r)*nu_r + D*nu_r + g(eta) = tau + tau_crossflow
%
% Inputs: L:   length
%          B:   beam
%          T:   draft
%          nu_r = [u-u_c, v-v_c, w-w_c, p, q, r]': Relative velocity vector
%          drag_model (Optional) : 'Hoerner' (default), 'cylinder'
%
% Output: tau_crossflow = [0 Yh Zh 0 Mh Nh]: 6-DOF cross-flow drag vector
```

```
rho = 1025; % Density of water
dx = L / 20; % Divide hull into 20 strips

switch drag_model
case 'Hoerner'
    % 2-D drag coefficient based on Hoerner's curve
    Cd_2D = Hoerner(B,T);
case 'cylinder'
    % 2D drag coefficient based on cylinder data from DNV-RP-C205
    Cd_2D = cylinderDrag(L,B,nu_r);
otherwise
    error('Unsupported drag model %s.',drag_model)
end

Yh = 0; Zh = 0; Mh = 0; Nh = 0;
for xL = -L/2:dx:L/2
    v_r = nu_r(2); % Relative sway velocity
    w_r = nu_r(3); % Relative heave velocity
    q = nu_r(5); % Pitch rate
    r = nu_r(6); % Yaw rate
    U_h = abs(v_r + xL * r) * (v_r + xL * r);
    U_v = abs(w_r + xL * q) * (w_r + xL * q);
    Yh = Yh - 0.5 * rho * T * Cd_2D * U_h * dx; % Sway force
    Zh = Zh - 0.5 * rho * T * Cd_2D * U_v * dx; % Heave force
    Mh = Mh - 0.5 * rho * T * Cd_2D * xL * U_v * dx; % Pitch moment
    Nh = Nh - 0.5 * rho * T * Cd_2D * xL * U_h * dx; % Yaw moment
end

tau_crossflow = [0 Yh Zh 0 Mh Nh]';
```

非线性横滚、俯仰和偏航阻尼

冲击阻力和横流阻力模型可以通过在横摇、纵摇和偏航中增加额外的非线性阻尼项来补充。这通常是必要的，以捕捉仅靠线性水动力阻尼无法很好表示的粘性和涡旋引起的阻尼效应，尤其是对于船舶和海上船只的横摇运动。

在偏航中，非线性阻尼项通常是必要的，以抵消由附加质量科里奥利力和向心力矩阵 $\mathbf{C}_A(\mathbf{v}_r)$ 引起的不稳定 Munk 力矩。在中等和高速前进时，Munk 力矩可能产生不稳定的偏航力矩，尤其对于细长船舶和水下航行器。

$$\begin{aligned}
 K &= K_p (1 + \kappa_4 |p|) p \\
 M &= M_q (1 + \kappa_5 |q|) q \\
 N &= N_r (1 + \kappa_6 |r|) r
 \end{aligned}$$

Coefficient	Mode	Units	Typical Range
κ_4	Roll	s	1–10
κ_5	Pitch	s	0.5–5
κ_6	Yaw	s	0.5–3



线性和非线性水动力阻尼

Hydrodynamic damping (Sections 6.1.1 and 6.1.3):

$$\mathbf{D}(\boldsymbol{\nu}_r) = - \begin{bmatrix} X_u^* & X_v & X_w & X_p & X_q & X_r \\ Y_u & Y_v & Y_w & Y_p & Y_q & Y_r \\ Z_u & Z_v & Z_w & Z_p & Z_q & Z_r \\ K_u & K_v & K_w & K_p^* & K_q & K_r \\ M_u & M_v & M_w & M_p & M_q^* & M_r \\ N_u & N_v & N_w & N_p & N_q & N_r^* \end{bmatrix}, \quad \begin{aligned} X_u^* &= e^{-k_u |u_r|} X_u + X_{|u|u} |u_r| \\ K_p^* &= K_p (1 + \kappa_4) |p| \\ M_q^* &= M_q (1 + \kappa_5) |q| \\ N_r^* &= N_r (1 + \kappa_6) |r| \end{aligned}$$

$$\mathbf{d}(\boldsymbol{\nu}_r) = -\frac{\rho C_d^{2D}}{2} \int_{-\frac{L}{2}}^{\frac{L}{2}} \begin{bmatrix} 0 \\ T(x) |v_r + xr|(v_r + xr) \\ T(x) |w_r + xq|(w_r + xq) \\ 0 \\ x T(x) |w_r + xq|(w_r + xq) \\ x T(x) |v_r + xr|(v_r + xr) \end{bmatrix} dx$$

使用二维阻力系数的条带理论横流阻力积分。

Alternatively, when maneuvering derivatives are unavailable, the linear damping coefficients $X_u, Y_v, Z_w, K_p, M_q, N_r$, etc. may be estimated from frequency-domain sea-keeping coefficients by power-based averaging (Section 5.4.2). This yields an equivalent constant linear damping matrix $\mathbf{B}_0^{\text{eq}}$, which can be combined with viscous damping contributions $\mathbf{B}_v$ to obtain $\mathbf{D} \approx \mathbf{B}_0^{\text{eq}} + \mathbf{B}_v$.

指数因子 $e^{-k_u |u_r|}$ 随速度增加而衰减线性突升阻尼系数 X_u , 以避免在包含二次阻尼 $X_{|u|u} u_r$ 时重复计算突升阻力。

名义上的滚转、俯仰和偏航线性阻尼导数通过系数 κ_4, κ_5 和 κ_6 的幅值相关阻尼得到增强。



摘要：基于第一性原理的六自由度机动模型

通常使用两种状态空间公式。第一种使用相对速度 $\mathbf{v}_r$ 作为状态，在明确建模洋流且测量相对速度（例如，通过 DVL）时很方便。第二种使用绝对速度 $\mathbf{v}$ 作为状态，在表面船舶中通常更受欢迎，因为 INS/GNSS 系统提供绝对速度的测量。

Independent of $\mathbf{v}_1 = [u, v, w]^T$

$$\underbrace{M_{RB}\dot{\mathbf{v}} + C_{RB}(\mathbf{v})\mathbf{v}}_{\text{Rigid-body forces}} + \underbrace{M_A\dot{\mathbf{v}}_r + C_A(\mathbf{v}_r)\mathbf{v}_r + D(\mathbf{v}_r)\mathbf{v}_r + d(\mathbf{v}_r)}_{\text{Hydrodynamic forces}}$$

$$+ \underbrace{g(\boldsymbol{\eta})}_{\text{Hydrostatic forces}} = \boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}}$$

$$\begin{aligned} M &= M_{RB} + M_A \\ C(\mathbf{v}_r) &= C_{RB}(\mathbf{v}_r) + C_A(\mathbf{v}_r) \end{aligned}$$

Relative velocity vector

$$\mathbf{v}_r = \mathbf{v} - \mathbf{v}_c$$

Irrotational ocean current

$$\mathbf{v}_c = \underbrace{[u_c, v_c, w_c, 0, 0, 0]^T}_{\mathbf{v}_c^b}$$

$$\mathbf{v}_c^b = [u_c, v_c, w_c]^T$$

$$\dot{\mathbf{v}}_c = \begin{bmatrix} -S(\boldsymbol{\omega}_{nb}^b)\mathbf{v}_c^b \\ \mathbf{0}_{3 \times 1} \end{bmatrix}$$

Relative-velocity state-space model:

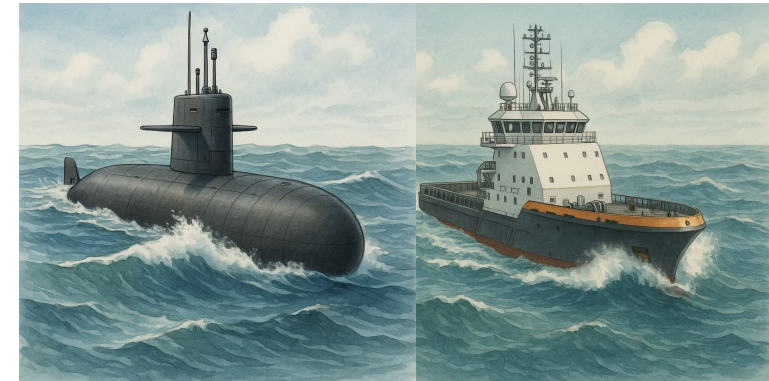
$$\dot{\boldsymbol{\eta}} = J_{\Theta}(\boldsymbol{\eta})(\mathbf{v}_r + \mathbf{v}_c)$$

$$\dot{\mathbf{v}}_r = M^{-1}(\boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}} - C(\mathbf{v}_r)\mathbf{v}_r - D(\mathbf{v}_r)\mathbf{v}_r - d(\mathbf{v}_r) - g(\boldsymbol{\eta}))$$

Absolute-velocity state-space model:

$$\dot{\boldsymbol{\eta}} = J_{\Theta}(\boldsymbol{\eta})\mathbf{v}$$

$$\dot{\mathbf{v}} = \dot{\mathbf{v}}_c + M^{-1}(\boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}} - C(\mathbf{v}_r)\mathbf{v}_r - D(\mathbf{v}_r)\mathbf{v}_r - d(\mathbf{v}_r) - g(\boldsymbol{\eta}))$$



$$\underbrace{M_{RB}\dot{\boldsymbol{\nu}} + C_{RB}(\boldsymbol{\nu})\boldsymbol{\nu}}_{\text{Rigid-body forces}} + \underbrace{M_A\dot{\boldsymbol{\nu}}_r + C_A(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + D(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{d}(\boldsymbol{\nu}_r)}_{\text{Hydrodynamic forces}} + \underbrace{\mathbf{g}(\boldsymbol{\eta})}_{\text{Hydrostatic forces}} = \boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}}$$

Rigid-body matrices (Section 3.3.2)

$$M_{RB} = \begin{bmatrix} m\mathbf{I}_3 & -m\mathbf{S}(\mathbf{r}_{bG}^b) \\ m\mathbf{S}(\mathbf{r}_{bG}^b) & \mathbf{I}_b^b \end{bmatrix}$$

$$C_{RB}(\boldsymbol{\nu}) = \mathbf{H}^\top(\mathbf{r}_{bG}^b) \begin{bmatrix} m\mathbf{S}(\boldsymbol{\nu}_2) & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & -\mathbf{S}(\mathbf{I}_b^b \boldsymbol{\nu}_2) \end{bmatrix} \mathbf{H}(\mathbf{r}_{bG}^b) \quad (\text{independent of } \boldsymbol{\nu}_1).$$

Added-mass matrices (Section 6.1.1)

$$M_A = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix} = - \begin{bmatrix} X_{\dot{u}} & X_{\dot{v}} & X_{\dot{w}} & X_{\dot{p}} & X_{\dot{q}} & X_{\dot{r}} \\ Y_{\dot{u}} & Y_{\dot{v}} & Y_{\dot{w}} & Y_{\dot{p}} & Y_{\dot{q}} & Y_{\dot{r}} \\ Z_{\dot{u}} & Z_{\dot{v}} & Z_{\dot{w}} & Z_{\dot{p}} & Z_{\dot{q}} & Z_{\dot{r}} \\ K_{\dot{u}} & K_{\dot{v}} & K_{\dot{w}} & K_{\dot{p}} & K_{\dot{q}} & K_{\dot{r}} \\ M_{\dot{u}} & M_{\dot{v}} & M_{\dot{w}} & M_{\dot{p}} & M_{\dot{q}} & M_{\dot{r}} \\ N_{\dot{u}} & N_{\dot{v}} & N_{\dot{w}} & N_{\dot{p}} & N_{\dot{q}} & N_{\dot{r}} \end{bmatrix}$$

$$C_A(\boldsymbol{\nu}_r) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & -\mathbf{S}(\mathbf{A}_{11}\boldsymbol{\nu}_{r,1} + \mathbf{A}_{12}\boldsymbol{\nu}_{r,2}) \\ -\mathbf{S}(\mathbf{A}_{11}\boldsymbol{\nu}_{r,1} + \mathbf{A}_{12}\boldsymbol{\nu}_{r,2}) & -\mathbf{S}(\mathbf{A}_{21}\boldsymbol{\nu}_{r,1} + \mathbf{A}_{22}\boldsymbol{\nu}_{r,2}) \end{bmatrix}$$

- Generalized position vector: $\boldsymbol{\eta} = [x^n, y^n, z^n, \phi, \theta, \psi]^\top$.
- Generalized velocity vector: $\boldsymbol{\nu} = [u, v, w, p, q, r]^\top$.
- Ocean current velocity vector: $\boldsymbol{\nu}_c = [u_c, v_c, w_c, 0, 0, 0]^\top$.
- Relative velocity vector: $\boldsymbol{\nu}_r = \boldsymbol{\nu} - \boldsymbol{\nu}_c$.

$$\begin{aligned} \boldsymbol{\nu}_{r,1} &= [u_r, v_r, w_r]^\top \\ \boldsymbol{\nu}_{r,2} &= \boldsymbol{\nu}_2 = [p, q, r]^\top \end{aligned} \quad \dot{\boldsymbol{\nu}}_c = \begin{bmatrix} -\mathbf{S}(\boldsymbol{\omega}_{nb}^b) \mathbf{v}_c^b \\ \mathbf{0}_{3 \times 1} \end{bmatrix}$$

Hydrodynamic damping (Sections 6.1.1 and 6.1.3)

$$D(\boldsymbol{\nu}_r) = - \begin{bmatrix} X_u^* & X_v & X_w & X_p & X_q & X_r \\ Y_u & Y_v & Y_w & Y_p & Y_q & Y_r \\ Z_u & Z_v & Z_w & Z_p & Z_q & Z_r \\ K_u & K_v & K_w & K_p^* & K_q & K_r \\ M_u & M_v & M_w & M_p & M_q^* & M_r \\ N_u & N_v & N_w & N_p & N_q & N_r^* \end{bmatrix}, \quad \begin{aligned} X_u^* &= e^{-k_u|u_r|} X_u + X_{|u|u}|u_r| \\ K_p^* &= K_p (1 + \kappa_4) |p| \\ M_q^* &= M_q (1 + \kappa_5) |q| \\ N_r^* &= N_r (1 + \kappa_6) |r| \end{aligned}$$

$$\mathbf{d}(\boldsymbol{\nu}_r) = -\frac{\rho C_d^{2D}}{2} \int_{-\frac{L}{2}}^{\frac{L}{2}} \begin{bmatrix} 0 \\ T(x) |v_r + xr|(v_r + xr) \\ T(x) |w_r + xq|(w_r + xq) \\ 0 \\ x T(x) |w_r + xq|(w_r + xq) \\ x T(x) |v_r + xr|(v_r + xr) \end{bmatrix} dx$$

Hydrostatic forces (Sections 4.1.1 and 4.2.1):

Neutrally buoyant submarines, ROVs and AUVs:

$$\mathbf{g}(\boldsymbol{\eta}) = [0, 0, 0, (z_G - z_B)W \cos(\theta) \sin(\phi), (z_G - z_B)W \sin(\theta), 0]^T$$

Ships and USVs:

$$\mathbf{g}(\boldsymbol{\eta}) \approx \mathbf{G}\boldsymbol{\eta}$$

where

$$\mathbf{G} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \rho g A_{wp} & 0 & -\rho g A_{wp} x_F & 0 \\ 0 & 0 & 0 & WGM_T & 0 & 0 \\ 0 & 0 & -\rho g A_{wp} x_F & 0 & \rho g A_{wp} x_F^2 + WGM_L & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

$$\underbrace{\mathbf{M}_{RB}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu}}_{\text{rigid-body forces}} + \underbrace{\mathbf{M}_A\dot{\boldsymbol{\nu}}_r + \mathbf{C}_A(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{D}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{d}(\boldsymbol{\nu}_r)}_{\text{hydrodynamic forces}} + \underbrace{\mathbf{g}(\boldsymbol{\eta})}_{\text{hydrostatic forces}} = \boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}}$$



特殊情况和简化机动模型

具有弱横向耦合的海上船舶纵向与横向操纵模型

Ships and USVs. For ships and USVs operating in waves, the added-mass and damping matrices are commonly approximated by constant equivalent matrices obtained from the frequency-dependent seakeeping coefficients:

$$\mathbf{M}_A = \mathbf{A}_0^{\text{eq}}, \quad \mathbf{D} = \mathbf{B}_0^{\text{eq}} + \mathbf{B}_v.$$

Underwater vehicles. For underwater vehicles operating below the wave-affected zone, the surrounding fluid may be regarded as unbounded. In this case, the hydrodynamic coefficients are independent of the wave-excitation frequency, so that

$$\begin{aligned} \mathbf{M}_A &= \mathbf{A}(\omega) = \text{constant} \quad \forall \omega, \\ \mathbf{D} &= \mathbf{B}_v, \quad \mathbf{B}(\omega) = \mathbf{0}. \end{aligned}$$



具有弱横向耦合的海上船舶纵向与横向操纵模型

假设 6.1（弱纵向-横向耦合）

The equations of motion are expressed in a body-fixed frame $\{b\}$ with origin at the CG. The craft is symmetric with respect to the xz -plane, and hydrodynamic coupling between longitudinal and lateral motions is neglected. The state vectors are

$$\begin{aligned}\boldsymbol{\eta}^{(\text{lon})} &= [x^n, z^n, \theta]^\top, & \boldsymbol{\nu}^{(\text{lon})} &= [u, w, q]^\top \\ \boldsymbol{\eta}^{(\text{lat})} &= [y^n, \phi, \psi]^\top, & \boldsymbol{\nu}^{(\text{lat})} &= [v, p, r]^\top.\end{aligned}$$

This approximation is often appropriate for slender marine vehicles operating near straight-line motion, including submarines, torpedo-shaped AUVs, and conventional monohull ships, where longitudinal and lateral dynamics are only weakly coupled.

接下来的幻灯片中的三自由度纵向和横向模型是基于以下假设和简化条件：

机体固定坐标原点：方程在以质心(CG)为原点的机体固定参考系中表示。这消除了与 x_G 、 y_G 和 z_G 相关的刚体耦合项。

对角线近似：在高保真机动模型中，通常保留非对角附加质量和阻尼项。然而，对于具有较弱交叉耦合的细长车辆，尤其是在只有有限水动力数据可用的情况下，通常将附加质量和线性阻尼矩阵近似为其对角线元素。在质心处表示方程可以减少被忽略的非对角项。

基于相对速度的纵向运动方程

$$\begin{aligned}
 \begin{bmatrix} \dot{x}^n \\ \dot{z}^n \\ \dot{\theta} \end{bmatrix} &= \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \left(\begin{bmatrix} u_r \\ w_r \\ q \end{bmatrix} + \begin{bmatrix} u_c \\ w_c \\ 0 \end{bmatrix} \right) \\
 \begin{bmatrix} m - X_{\dot{u}} & 0 & 0 \\ 0 & m - Z_{\dot{w}} & 0 \\ 0 & 0 & I_y - M_{\dot{q}} \end{bmatrix} \begin{bmatrix} \dot{u}_r \\ \dot{w}_r \\ \dot{q} \end{bmatrix} \\
 &+ \begin{bmatrix} 0 & 0 & (m - Z_{\dot{w}})w_r \\ 0 & 0 & -(m - X_{\dot{u}})u_r \\ -(m - Z_{\dot{w}})w_r & (m - X_{\dot{u}})u_r & 0 \end{bmatrix} \begin{bmatrix} u_r \\ w_r \\ q \end{bmatrix} \\
 &+ \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & \rho g A_{wp} & -\rho g A_{wp} x_F \\ 0 & -\rho g A_{wp} x_F & \rho g A_{wp} x_F^2 + WGM_L \end{bmatrix}}_{\text{For submerged vehicles, replace by } [0, -z_B W \sin(\theta), 0]^T} \begin{bmatrix} x^n \\ z^n \\ \theta \end{bmatrix} \\
 &+ \begin{bmatrix} -e^{-k_u |u_r|} X_u - X_{|u|u} |u_r| & 0 & 0 \\ 0 & -Z_w & 0 \\ 0 & 0 & -M_q (1 + \kappa_5 |q|) \end{bmatrix} \begin{bmatrix} u_r \\ w_r \\ q \end{bmatrix} \\
 &- \underbrace{\frac{\rho C_d^{2D}}{2} \int_{-\frac{L}{2}}^{\frac{L}{2}} \begin{bmatrix} 0 \\ T(x) |w_r + xq| (w_r + xq) \\ x T(x) |w_r + xq| (w_r + xq) \end{bmatrix} dx}_{\text{Optional crossflow drag}} = \begin{bmatrix} \tau_1 \\ \tau_3 \\ \tau_5 \end{bmatrix} + \begin{bmatrix} \tau_{\text{wind},1} \\ \tau_{\text{wind},3} \\ \tau_{\text{wind},5} \end{bmatrix} + \begin{bmatrix} \tau_{\text{wave},1} \\ \tau_{\text{wave},3} \\ \tau_{\text{wave},5} \end{bmatrix}
 \end{aligned}$$



基于相对速度的横向运动方程

$$\begin{bmatrix} \dot{y}^n \\ \dot{\phi} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} \cos(\psi) & 0 & 0 \\ 0 & 1 & \sin(\phi) \tan(\theta) \\ 0 & 0 & \cos(\phi) / \cos(\theta) \end{bmatrix} \left(\begin{bmatrix} v_r \\ p \\ r \end{bmatrix} + \begin{bmatrix} v_c \\ 0 \\ 0 \end{bmatrix} \right)$$

$$\begin{bmatrix} m - Y_{\dot{v}} & 0 & 0 \\ 0 & I_x - K_{\dot{p}} & 0 \\ 0 & 0 & I_z - N_{\dot{r}} \end{bmatrix} \begin{bmatrix} \dot{v}_r \\ \dot{p} \\ \dot{r} \end{bmatrix} + \begin{bmatrix} 0 & -(m - Z_{\dot{w}})w_r & (m - X_{\dot{u}})u_r \\ (m - Z_{\dot{w}})w_r & 0 & 0 \\ -(m - X_{\dot{u}})u_r & 0 & 0 \end{bmatrix} \begin{bmatrix} v_r \\ p \\ r \end{bmatrix}$$

$$+ \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & WGM_T & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{\text{For submerged vehicles, replace by } [0, -z_B W \sin(\phi), 0]^T} \begin{bmatrix} y^n \\ \phi \\ \psi \end{bmatrix}$$

For submerged vehicles,
replace by $[0, -z_B W \sin(\phi), 0]^T$

$$+ \begin{bmatrix} -Y_v & 0 & 0 \\ 0 & -K_p(1 + \kappa_4|p|) & 0 \\ 0 & 0 & -N_r(1 + \kappa_6|r|) \end{bmatrix} \begin{bmatrix} v_r \\ p \\ r \end{bmatrix}$$

$$- \underbrace{\frac{\rho C_d^2 D}{2} \int_{-\frac{L}{2}}^{\frac{L}{2}} \begin{bmatrix} T(x) |v_r + xr|(v_r + xr) \\ 0 \\ x T(x) |v_r + xr|(v_r + xr) \end{bmatrix} dx}_{\text{Optional crossflow drag}} = \begin{bmatrix} \tau_2 \\ \tau_4 \\ \tau_6 \end{bmatrix} + \begin{bmatrix} \tau_{\text{wind},2} \\ \tau_{\text{wind},4} \\ \tau_{\text{wind},6} \end{bmatrix} + \begin{bmatrix} \tau_{\text{wave},2} \\ \tau_{\text{wave},4} \\ \tau_{\text{wave},6} \end{bmatrix}$$



具有左右对称性的船舶和无人水面船水平面机动模型



Assumption 6.2 (Horizontal-plane motion)

Assume port–starboard symmetry such that $y_G = 0$, and choose the body-fixed frame $\{b\}$ so that the CG is located at $(x_G, 0, 0)$. Furthermore, assume that heave, roll, and pitch motions are small and may be neglected. Under these assumptions, the craft motion is approximated by the horizontal-plane states (x^n, y^n, ψ) and (u, v, r) .

基于相对速度的非线性运动方程

$$\underbrace{M_{RB}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu}}_{\text{rigid-body forces}} + \underbrace{M_A\dot{\boldsymbol{\nu}}_r + \mathbf{C}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{D}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r}_{\text{hydrodynamic forces}} = \boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}}$$

$$\dot{\boldsymbol{\eta}} = \mathbf{R}(\psi)\boldsymbol{\nu}$$

The nonlinear hydrodynamic derivatives in sway and yaw can also be approximated by cross-flow drag integrals

$$\begin{bmatrix} Y_{|v|v}|v_r| + Y_{|r|v}|r| & Y_{|v|r}|v_r| + Y_{|r|r}|r| \\ N_{|v|v}|v_r| + N_{|r|v}|r| & N_{|v|r}|v_r| + N_{|r|r}|r| \end{bmatrix} \begin{bmatrix} v_r \\ r \end{bmatrix} \approx -\frac{\rho C_d^{2D}}{2} \int_{-L/2}^{L/2} \begin{bmatrix} T(x)|v_r + xr|(v_r + xr) \\ xT(x)|v_r + xr|(v_r + xr) \end{bmatrix} dx$$



$$\mathbf{M}_{RB} = \begin{bmatrix} m & 0 & 0 \\ 0 & m & mx_G \\ 0 & mx_G & I_z \end{bmatrix}, \quad \mathbf{M}_A = -\begin{bmatrix} X_{\dot{u}} & 0 & 0 \\ 0 & Y_{\dot{v}} & Y_{\dot{r}} \\ 0 & N_{\dot{v}} & N_{\dot{r}} \end{bmatrix}$$

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} 0 & 0 & -m(x_G r + v) \\ 0 & 0 & mu \\ m(x_G r + v) & -mu & 0 \end{bmatrix}$$

$$\mathbf{C}_A(\boldsymbol{\nu}_r) = \begin{bmatrix} 0 & 0 & Y_{\dot{v}}v_r + Y_{\dot{r}}r \\ 0 & 0 & -X_{\dot{u}}u_r \\ -Y_{\dot{v}}v_r - Y_{\dot{r}}r & X_{\dot{u}}u_r & 0 \end{bmatrix}$$

$$\mathbf{D}(\boldsymbol{\nu}_r) = -\begin{bmatrix} e^{-k_u|u_r|}X_u & 0 & 0 \\ 0 & Y_v & Y_r \\ 0 & N_v & N_r \end{bmatrix}$$

$$-\begin{bmatrix} X_{|u|u}|u_r| & 0 & 0 \\ 0 & Y_{|v|v}|v_r| + Y_{|r|v}|r| & Y_{|v|r}|v_r| + Y_{|r|r}|r| \\ 0 & N_{|v|v}|v_r| + N_{|r|v}|r| & N_{|v|r}|v_r| + N_{|r|r}|r| \end{bmatrix}$$

具有左右对称性的船舶和无人水面船水平面机动模型

由于 $\mathbf{C}_{RB}(\mathbf{v})$ 是参数 独立于线速度的etrized, 由性质得出

而 10.1 那个

$$\mathbf{M}\dot{\mathbf{v}}_r + \mathbf{C}(\mathbf{v}_r)\mathbf{v}_r + \mathbf{D}(\mathbf{v}_r)\mathbf{v}_r = \boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}}$$

$$\begin{aligned}\mathbf{M} &= \mathbf{M}_A + \mathbf{M}_{RB} \\ \mathbf{C}(\mathbf{v}_r) &= \mathbf{C}_A(\mathbf{v}_r) + \mathbf{C}_{RB}(\mathbf{v}_r).\end{aligned}$$

水平面运动的状态空间模型

$$\begin{aligned}\dot{\boldsymbol{\eta}} &= \mathbf{R}(\psi) \left(\mathbf{v}_r + \begin{bmatrix} u_c \\ v_c \\ 0 \end{bmatrix} \right) \\ \dot{\mathbf{v}}_r &= \mathbf{M}^{-1}(\boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}} - \mathbf{C}(\mathbf{v}_r)\mathbf{v}_r - \mathbf{D}(\mathbf{v}_r)\mathbf{v}_r)\end{aligned}$$

$$\begin{aligned}u_c &= V_c \cos(\beta_{V_c} - \psi) \\ v_c &= V_c \sin(\beta_{V_c} - \psi)\end{aligned}$$

$$\begin{aligned}\dot{\boldsymbol{\eta}} &= \mathbf{R}(\psi)\mathbf{v} \\ \dot{\mathbf{v}} &= \begin{bmatrix} rv_c \\ -ru_c \\ 0 \end{bmatrix} + \mathbf{M}^{-1}(\boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}} - \mathbf{C}(\mathbf{v}_r)\mathbf{v}_r - \mathbf{D}(\mathbf{v}_r)\mathbf{v}_r) \\ \text{where } \mathbf{v}_r &= \mathbf{v} - [u_c, v_c, 0]^\top.\end{aligned}$$

$$\begin{aligned}\dot{u}_c &= rV_c \sin(\beta_{V_c} - \psi) = rv_c \\ \dot{v}_c &= -rV_c \cos(\beta_{V_c} - \psi) = -ru_c\end{aligned}$$

具有弱横向耦合的海上船舶纵向与横向操纵模型

MSS 工具箱中包含了若干线性和非线性机动模型。

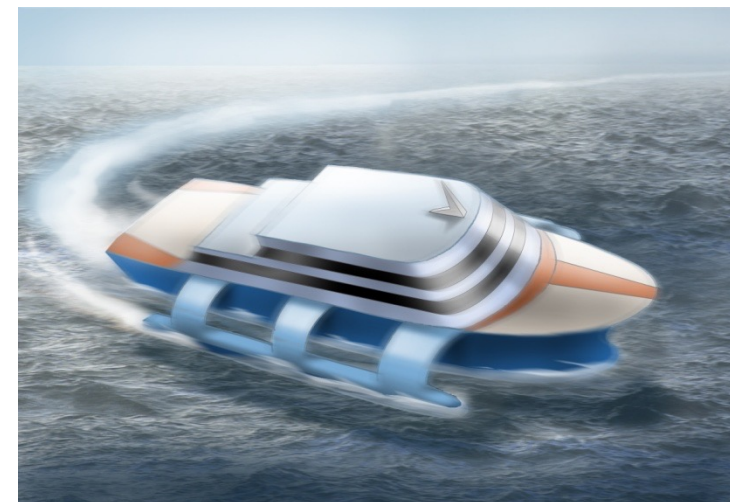
Matlab/GNU Octave:

Several MSS maneuvering models are located in the toolbox catalog
../MSS/CRAFT/.

```
[xdot,U] = container(x,tau)    % Container ship L = 175 m
[xdot,U] = Lcontainer(x,tau)   % Linearized container ship
[xdot,U] = mariner(x,ui,U0)    % Mariner class vessel L = 161 m
[xdot,U] = navalvessel(x,tau)  % Naval vessel L = 51.5 m
[xdot,U] = tanker(x,ui)        % Tanker L = 304.8 m
[xdot,U,M,B_p] = otter(x,n,mp,rp,V_c,beta_c) % Otter USV L = 2 m
```

The models can be simulated under feedback control using the user editable scripts

Sim<model name>.m



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MSS 工具箱

C在BODY和NED中计算海流速度

我们假设电流在NED中是无旋的且恒定。因此,

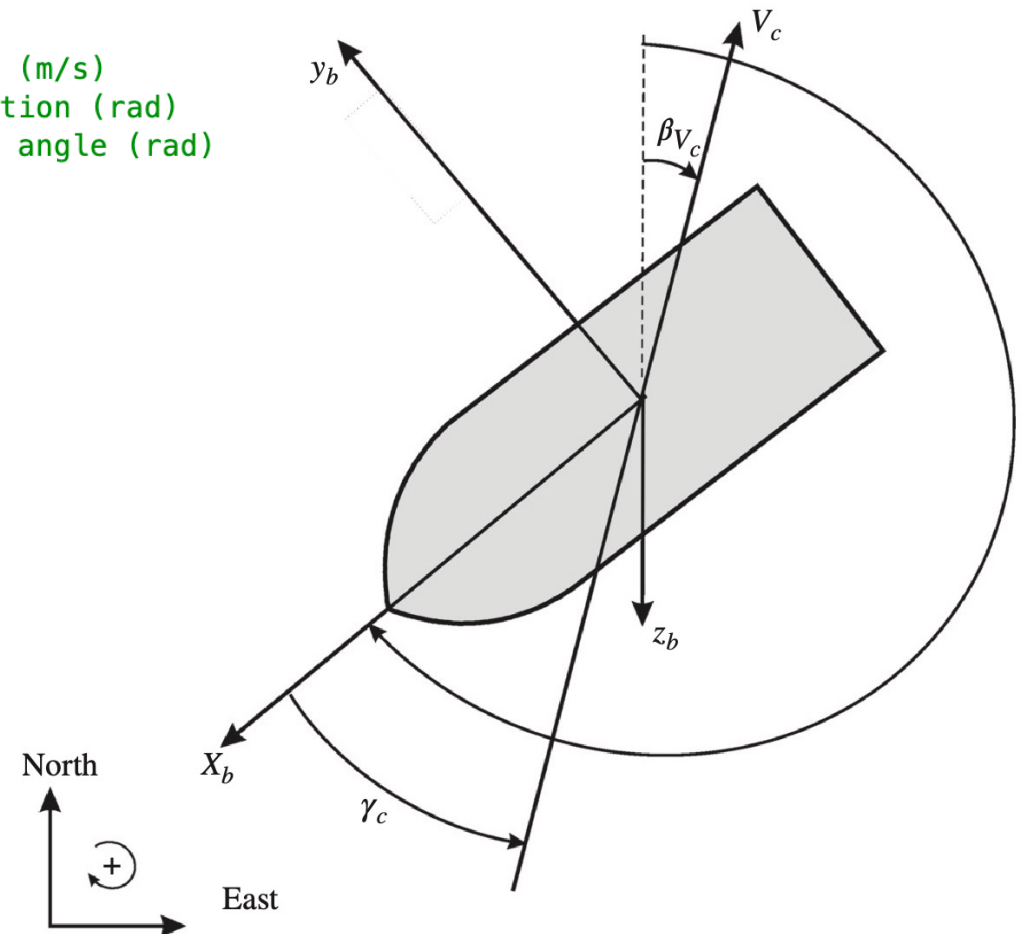
```
% constant 2-D ocean current expressed in NED
Vc = 1; % current speed (m/s)
betaVc = deg2rad(30); % current direction (rad)
psi = deg2rad(10); % craft heading angle (rad)
```

```
% current velocities expressed in NED
vc_n = [Vc * cos(betaVc) Vc * sin(betaVc) 0]'
```

```
% current velocities expressed in BODY
vc_b = Rzyx(0,0,psi)' * vc_n
```

```
vc_n =
    0.8660
    0.5000
         0
vc_b =
    0.9397
    0.3420
         0
```

```
>> help Rzyx
R = Rzyx(phi,theta,psi) computes the Euler angle
rotation matrix R in SO(3) using the zyx convention
```



适用于船舶和无人水面艇的三自由度线性机动模型

线性操纵方程（纵摇-横摇-偏航）

$$\underbrace{(M_{RB} + M_A)}_M \dot{\nu}_r + \underbrace{(C_{RB} + C_A + D)}_N \nu_r = \tau + \tau_{\text{wind}} + \tau_{\text{wave}}$$



$$\begin{bmatrix} m - X_{\dot{u}} & 0 & 0 \\ 0 & m - Y_{\dot{v}} & mx_G - Y_{\dot{r}} \\ 0 & mx_G - N_{\dot{v}} & I_z - N_{\dot{r}} \end{bmatrix} \begin{bmatrix} \dot{u}_r \\ \dot{v}_r \\ \dot{r} \end{bmatrix} + \begin{bmatrix} -X_u & 0 & 0 \\ 0 & -Y_v & (m - X_{\dot{u}})U_0 - Y_r \\ 0 & (X_{\dot{u}} - Y_{\dot{v}})U_0 - N_v & (mx_G - Y_{\dot{r}})U_0 - N_r \end{bmatrix} \begin{bmatrix} u_r \\ v_r \\ r \end{bmatrix} = \begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_6 \end{bmatrix}$$

For a rudder operating in the propeller slipstream,

$$\tau_1 = -X_{\delta\delta}\delta^2 + (1 - t)T_p, \quad \tau_2 = -Y_{\delta}\delta, \quad \tau_6 = -N_{\delta}\delta$$

$$C_{RB} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & mU_0 \\ 0 & 0 & mx_GU_0 \end{bmatrix}, \quad C_A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -X_{\dot{u}}U_0 \\ 0 & (X_{\dot{u}} - Y_{\dot{v}})U_0 & -Y_{\dot{r}}U_0 \end{bmatrix}$$

$$D = - \begin{bmatrix} X_u & 0 & 0 \\ 0 & Y_v & Y_r \\ 0 & N_v & N_r \end{bmatrix}$$

U_0 is the cruise speed
用于线性化



适用于船舶和无人水面艇的三自由度线性机动模型

将纵荡-横荡-偏航模型解耦为纵荡子系统和横荡-偏航子系统:

浪涌子系统

$$(m - X_{\dot{u}})\dot{u}_r - X_u u_r = X_{\delta\delta}\delta^2 + (1 - t_p)T_p$$

Sway-Yaw Subsystem

$$X_u = \frac{8A_1}{3\pi} X_{|u|u}$$

$$M\dot{\nu}_r + N\nu_r = b\delta$$

$$M = \begin{bmatrix} m - Y_{\dot{v}} & mx_G - Y_{\dot{r}} \\ mx_G - N_{\dot{v}} & I_z - N_{\dot{r}} \end{bmatrix}$$

$$N = \begin{bmatrix} -Y_v & (m - X_{\dot{u}})U_0 - Y_r \\ (X_{\dot{u}} - Y_{\dot{v}})U_0 - N_v & (mx_G - Y_{\dot{r}})U_0 - N_r \end{bmatrix}$$

$$b = \begin{bmatrix} -Y_{\delta} \\ -N_{\delta} \end{bmatrix}.$$

Notice: $C_A(v)$ includes the famous destabilizing Munk moment (from aerodynamics) and some other C_A -terms



用于动态定位 (DP) 的三自由度线性模型

DP 模型是在低速运行假设下推导的 (通常最高可达 2 米/秒)，因此二次科里奥利力和向心力项以及非线性阻尼力相对于线性阻尼项可以忽略不计。

运动学非线性

$$\mathbf{R}(\psi(t)) := \mathbf{R}(t) \quad (\text{与旋转矩阵相关的非线性可以通过将偏航角视为已测量来消除})$$

用于DP控制器-观测器设计的线性时变 (LTV) 模型

$$\begin{aligned} \dot{\eta} &= \mathbf{R}(t)\nu \\ \mathbf{M}\dot{\nu} + \mathbf{D}\nu &= \mathbf{R}^\top(t)\mathbf{b} + \boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}} \\ \dot{\mathbf{b}} &= \mathbf{0} \quad (\text{The bias } \mathbf{b} \text{ is constant in NED}) \end{aligned}$$

$$\boldsymbol{\tau} = \mathbf{B}\mathbf{u} \quad (\text{Control forces and moments})$$

Matlab / GNU Octave:

The MSS DP models are located in the catalog `../MSS/CRAFT/SHIPS/`.

```
[xdot,U,M] = osv(x,ui,Vc,betaVc) % Supply vessel L = 83.0 m
[xdot,U,M,D] = supply(x, tau)    % Supply vessel L = 76.2 m
```

The models can be simulated under feedback control using the user editable scripts

`Sim<model name>.m`

线性时变 (LTV) 状态空间模型

$$\dot{\mathbf{x}} = \underbrace{\begin{bmatrix} \mathbf{0}_{3 \times 3} & \mathbf{R}(t) & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & -\mathbf{M}^{-1}\mathbf{D} & \mathbf{M}^{-1}\mathbf{R}^\top(t) \\ \mathbf{0}_{3 \times 3} & \mathbf{0}_{3 \times 3} & \mathbf{0}_{3 \times 3} \end{bmatrix}}_{\mathbf{A}(t)} \mathbf{x} + \underbrace{\begin{bmatrix} \mathbf{0}_{3 \times r} \\ \mathbf{M}^{-1}\mathbf{B} \\ \mathbf{0}_{3 \times r} \end{bmatrix}}_{\mathbf{B}} \mathbf{u} + \underbrace{\begin{bmatrix} \mathbf{0}_{3 \times 3} \\ \mathbf{M}^{-1} \\ \mathbf{0}_{3 \times 3} \end{bmatrix}}_{\mathbf{E}} \mathbf{w}$$

$$\mathbf{x} = [\eta^\top, \nu^\top, \mathbf{b}^\top]^\top$$

$$\mathbf{w} = \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}}$$



经典机动模型

阿布科维茨（1964）机动模型

数据驱动的机动建模

以前的操作模型是从第一性原理推导出来的：

- 浪涌抗性
- 横向流阻
- 水动力力模型

一种替代方法是直接拟合实验机动数据。

阿布考维茨的想法（1964）

用一个截断的表示浪涌力、摇摆力和偏航力矩
关于直线运动的泰勒级数展开：

$$\mathbf{M}_{RB}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} = \boldsymbol{\tau}_{RB}$$

$$\boldsymbol{\tau}_{RB} = [X(\mathbf{x}), Y(\mathbf{x}), N(\mathbf{x})]^\top$$

关键假设

1. 关于 $(u=U_0)$ 的三阶泰勒展开。
2. 加速度项仅以线性方式出现。
3. 强制实施左舷-右舷对称。
4. 忽略加速度-速度耦合项。

$$\mathbf{x} = [u, v, r, \dot{u}, \dot{v}, \dot{r}, \delta]^\top$$

$$X(\mathbf{x}) \approx X(\mathbf{x}_0) + \sum_{i=1}^n \left(\frac{\partial X(\mathbf{x})}{\partial x_i} \Big|_{\mathbf{x}_0} \Delta x_i + \frac{1}{2} \frac{\partial^2 X(\mathbf{x})}{(\partial x_i)^2} \Big|_{\mathbf{x}_0} \Delta x_i^2 + \frac{1}{6} \frac{\partial^3 X(\mathbf{x})}{(\partial x_i)^3} \Big|_{\mathbf{x}_0} \Delta x_i^3 \right)$$

$$Y(\mathbf{x}) \approx Y(\mathbf{x}_0) + \sum_{i=1}^n \left(\frac{\partial Y(\mathbf{x})}{\partial x_i} \Big|_{\mathbf{x}_0} \Delta x_i + \frac{1}{2} \frac{\partial^2 Y(\mathbf{x})}{(\partial x_i)^2} \Big|_{\mathbf{x}_0} \Delta x_i^2 + \frac{1}{6} \frac{\partial^3 Y(\mathbf{x})}{(\partial x_i)^3} \Big|_{\mathbf{x}_0} \Delta x_i^3 \right)$$

$$N(\mathbf{x}) \approx Z(\mathbf{x}_0) + \sum_{i=1}^n \left(\frac{\partial N(\mathbf{x})}{\partial x_i} \Big|_{\mathbf{x}_0} \Delta x_i + \frac{1}{2} \frac{\partial^2 N(\mathbf{x})}{(\partial x_i)^2} \Big|_{\mathbf{x}_0} \Delta x_i^2 + \frac{1}{6} \frac{\partial^3 N(\mathbf{x})}{(\partial x_i)^3} \Big|_{\mathbf{x}_0} \Delta x_i^3 \right)$$

$$\Delta \mathbf{x} = \mathbf{x} - \mathbf{x}_0 = [\Delta x_1, \Delta x_2, \dots, \Delta x_7]^\top$$

$$\mathbf{x}_0 = [U, 0, 0, 0, 0, 0, 0]^\top$$

阿布科维茨（1964）机动模型

$$\mathbf{M}_{RB}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} = \boldsymbol{\tau}_{RB}$$

$$\boldsymbol{\tau}_{RB} = [X(\mathbf{x}), Y(\mathbf{x}), N(\mathbf{x})]^\top$$

泰勒级数展开

$$\begin{aligned} X = & X^* + X_{\dot{u}}\dot{u} + X_u\Delta u + X_{uu}\Delta u^2 + X_{uuu}\Delta u^3 + X_{vv}v^2 + X_{rr}r^2 + X_{\delta\delta}\delta^2 \\ & + X_{rv\delta}rv\delta + X_{r\delta}r\delta + X_{v\delta}v\delta + X_{vvu}v^2\Delta u + X_{rru}r^2\Delta u + X_{\delta\delta u}\delta^2\Delta u \\ & + X_{rvu}rvu + X_{r\delta u}r\delta\Delta u + X_{v\delta u}v\delta\Delta u \end{aligned}$$

$$\begin{aligned} Y = & Y^* + Y_u\Delta u + Y_{uu}\Delta u^2 + Y_rr + Y_vv + Y_{\dot{r}}\dot{r} + Y_{\dot{v}}\dot{v} + Y_{\delta}\delta + Y_{rrr}r^3 + Y_{vvv}v^3 \\ & + Y_{\delta\delta\delta}\delta^3 + Y_{rr\delta}r^2\delta + Y_{\delta\delta r}\delta^2r + Y_{rrv}r^2v + Y_{vvv}v^2r + Y_{\delta\delta v}\delta^2v + Y_{vv\delta}v^2\delta \\ & + Y_{\delta vr}\delta vr + Y_{vu}v\Delta u + Y_{vu}v\Delta u^2 + Y_{ru}r\Delta u + Y_{ruu}r\Delta u^2 + Y_{\delta u}\delta\Delta u \\ & + Y_{\delta uu}\delta\Delta u^2 \end{aligned}$$

$$\begin{aligned} N = & N^* + N_u\Delta u + N_{uu}\Delta u^2 + N_rr + N_vv + N_{\dot{r}}\dot{r} + N_{\dot{v}}\dot{v} + N_{\delta}\delta + N_{rrr}r^3 \\ & + N_{vvv}v^3 + N_{\delta\delta\delta}\delta^3 + N_{rr\delta}r^2\delta + N_{\delta\delta r}\delta^2r + N_{rrv}r^2v + N_{vvv}v^2r + N_{\delta\delta v}\delta^2v \\ & + N_{vv\delta}v^2\delta + N_{\delta vr}\delta vr + N_{vu}v\Delta u + N_{vu}v\Delta u^2 + N_{ru}r\Delta u + N_{ruu}r\Delta u^2 \\ & + N_{\delta u}\delta\Delta u + N_{\delta uu}\delta\Delta u^2 \end{aligned}$$



二阶模量模型（费佳耶夫斯基-索博列夫模型）

使用二阶模函数来表示非线性耗散力和力矩的想法源自 Fedyaevsky 和 Sobolev（1963 年）的工作。该模型特别适用于中高速的机动，在此情况下，水动力载荷以二次横流阻力和推进损失为主。Norrbinn（1970 年）采用了这种建模理念，提出了一种基于水动力力和力矩模函数表示的非线性机动模型。

$$M\dot{\nu}_r + C(\nu_r)\nu_r + D(\nu_r)\nu_r = \tau + \tau_{\text{wind}} + \tau_{\text{wave}}$$

$$\begin{aligned} & C_A(\nu_r)\nu_r + D(\nu_r)\nu_r \\ &= \begin{bmatrix} Y_{\dot{v}}v_r r + Y_{\dot{r}}r^2 \\ -X_{\dot{u}}u_r r \\ (X_{\dot{u}} - Y_{\dot{v}})u_r v_r - Y_{\dot{r}}u_r r \end{bmatrix} \left(\text{alternatively} \begin{bmatrix} X_{vr}v_r r + X_{rr}r^2 \\ Y_{ur}u_r r \\ N_{uv}u_r v_r + N_{ur}u_r r \end{bmatrix} \right) \\ &+ \begin{bmatrix} -X_{|u|u}|u_r|u_r \\ -Y_{|v|v}|v_r|v_r - Y_{|v|r}|v_r|r - Y_{v|r}|v_r|r| - Y_{|r|r}|r|r| \\ -N_{|v|v}|v_r|v_r - N_{|v|r}|v_r|r - N_{v|r}|v_r|r| - N_{|r|r}|r|r| \end{bmatrix} \\ &= \begin{bmatrix} -X_{|u|u}|u_r|u_r + Y_{\dot{v}}v_r r + Y_{\dot{r}}r^2 \\ -X_{\dot{u}}u_r r - Y_{|v|v}|v_r|v_r - Y_{|v|r}|v_r|r - Y_{v|r}|v_r|r| - Y_{|r|r}|r|r| \\ (X_{\dot{u}} - Y_{\dot{v}})u_r v_r - Y_{\dot{r}}u_r r - N_{|v|v}|v_r|v_r \\ -N_{|v|r}|v_r|r - N_{v|r}|v_r|r| - N_{|r|r}|r|r| \end{bmatrix} \end{aligned}$$

¹The $C_A(\nu_r)\nu_r$ terms may alternatively be written as $X_{vr}v_r r$, $X_{rr}r^2$, $Y_{ur}u_r r$, $N_{uv}u_r v_r$, and $N_{ur}u_r r$. If these coefficients are obtained experimentally, they generally contain viscous contributions in addition to the potential-flow coefficients $Y_{\dot{v}}$, $X_{\dot{u}}$, and $Y_{\dot{r}}$.



二阶模量模型 (Blanke, 1981)

$$M\dot{\nu}_r + C(\nu_r)\nu_r + D(\nu_r)\nu_r = \tau + \tau_{\text{wind}} + \tau_{\text{wave}}$$

$$C_A(\nu_r) = \begin{bmatrix} 0 & 0 & Y_{\dot{v}}v_r + Y_{\dot{r}}r \\ 0 & 0 & -X_{\dot{u}}u_r \\ -Y_{\dot{v}}v_r - Y_{\dot{r}}r & X_{\dot{u}}u_r & 0 \end{bmatrix}$$

$$D(\nu_r) = \begin{bmatrix} -X_{|u|u}|u_r| & 0 & 0 \\ 0 & -Y_{|v|v}|v_r| - Y_{|r|v}|r| & -Y_{|v|r}|v_r| - Y_{|r|r}|r| \\ 0 & -N_{|v|v}|v_r| - N_{|r|v}|r| & -N_{|v|r}|v_r| - N_{|r|r}|r| \end{bmatrix}$$

For large ships, Blanke (1981) suggested a simplified damping model which neglects the $|r|$ -dependent terms. Hence,

$$C_A(\nu_r) + D(\nu_r) = \begin{bmatrix} -X_{|u|u}|u_r| & 0 & Y_{\dot{v}}v_r + Y_{\dot{r}}r \\ 0 & -Y_{|v|v}|v_r| & -X_{\dot{u}}u_r - Y_{|v|r}|v_r| \\ -Y_{\dot{v}}v_r - Y_{\dot{r}}r & X_{\dot{u}}u_r - N_{|v|v}|v_r| & -N_{|v|r}|v_r| \end{bmatrix}$$

Blanke (1981) 提出了诺尔宾模型的简化版本，仅保留用于舵效和推进损失预测的最重要项。该模型可以被解释为横流阻力积分的二阶近似。



MMG标准操纵模型

船舶机动建模组（MMG）标准模型，由日本MMG委员会开发，后来由Yasukawa和Yoshimura（2015）标准化，是一个用于船舶纵摇、横摇和偏航的经验三自由度机动模型。与二阶模量模型和Abkowitz模型不同，后者的水动力作用力和偏航力矩直接由船体固定速度的多项式或模量函数表示，MMG模型将总力和力矩分解为船体、螺旋桨和舵的单独贡献。这种分解赋予模型明确的物理解释，并使得将固定模型试验与螺旋桨和舵的经验公式结合成为可能。

$$M\dot{\boldsymbol{\nu}}_r + \mathbf{C}_{RB}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r = \boldsymbol{\tau}_H + \boldsymbol{\tau}_P + \boldsymbol{\tau}_R + \boldsymbol{\tau}_{\text{env}}$$

$$\begin{bmatrix} X \\ Y \\ N \end{bmatrix} = \begin{bmatrix} X_H + X_P + X_R \\ Y_H + Y_R \\ N_H + N_R \end{bmatrix}$$

船体力和力矩的贡献由非线性水动力导数表示。一个常见的MMG表示方法是

$$\begin{aligned} X_H &= X_{uu}u_r^2 + X_{vv}v_r^2 + X_{vr}v_r r + X_{rr}r^2 \\ Y_H &= Y_v v_r + Y_r r + Y_{|v|v}|v_r|v_r + Y_{|r|r}|r|r + Y_{vr}v_r r \\ N_H &= N_v v_r + N_r r + N_{|v|v}|v_r|v_r + N_{|r|r}|r|r + N_{vr}v_r r \end{aligned}$$

推进力（第9.1节）

$$X_P = (1 - t_p) T_p, \quad T_p = \rho D_p^4 K_T(J_a) |n_p| n_p$$

舵力（第9.5节）

$$\begin{aligned} X_R &= -(1 - t_R) F_N \sin(\delta) \\ Y_R &= -(1 + a_H) F_N \cos(\delta) \\ N_R &= -(x_R + a_H x_H) F_N \cos(\delta) \end{aligned}$$

$$F_N = \frac{1}{2} \rho A_R f_\alpha U_R^2 \sin(\alpha_R)$$

章节目标 - 再访

- 通过将刚体动力学与流体动力附加质量、阻尼、静水恢复力和外力相结合，制定基于第一性原理的六自由度机动模型。
- 解释刚体和附加质量科里奥利-向心矩阵在机体固定坐标系中是如何产生的，并说明它们的斜对称性特性。
- 解释在纵荡、横荡和偏航中何时适合使用零频潜势系数，为什么必须添加粘性阻尼，以及为什么该近似在耦合6自由度运动中失效。
- 应用基于功率的谱平均方法来计算6自由度机动模型的海况相关恒定等效附加质量和阻尼矩阵。
- 使用 ITTC 冲程阻力、横向流拖力以及额外的横摇、纵倾和偏航阻尼来建模线性和非线性流体阻尼。
- 使用相对速度包含恒定无旋转的海洋洋流，并用相对速度或绝对速度作为状态来制定动力学。
- 推导并比较简化和经典的机动模型，包括纵横向、水平面三自由度、动态定位、Abkowitz、二阶模量和MMG模型。

$$\underbrace{M_{RB}\dot{\nu} + C_{RB}(\nu)\nu}_{\text{rigid-body forces}} + \underbrace{M_A\dot{\nu}_r + C_A(\nu_r)\nu_r + D(\nu_r)\nu_r + d(\nu_r)}_{\text{hydrodynamic forces}} + \underbrace{g(\eta)}_{\text{hydrostatic forces}} = \tau + \tau_{\text{wind}} + \tau_{\text{wave}}$$